A Data-Driven, Component-Level Projection Framework for 21st-Century Sea-Level Rise

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Abstract

Every centimeter of sea-level rise exposes millions more people to annual coastal flooding, yet projections of sea level change come with large uncertainties that arise in part from model structure and parameters, and are rarely quantified in terms of their source. Here, we construct a data-driven sea-level projection framework by calibrating parsimonious physical models for individual components of sea-level change against decades-long independent observational records. We call this framework the Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models (SPECTRUM). As model validation, we force SPECTRUM with observed global mean surface temperatures (GMST) and find that the calibrated components sum to match global mean sea level (GMSL) rise since 1993 as measured with satellite altimetry. Hindcasts that start in 2025 and are forced with observed GMST match observed GMSL within uncertainty back to at least 1950 and approximately 1.2 °C of warming, suggesting forecasting skill for the GMST-sensitive components until the end of the 21st century. Our model cannot constrain the probability of mass loss from West Antarctica, so we treat it with a 'storyline' as in previous studies. The lower bound of the storyline is constrained by extrapolating the observed mass loss trend while the upper bound is constrained by structured expert judgment. In both cases, the dynamics of West Antarctic glaciers are the largest source of uncertainty, accounting for more than 90% of the sea-level change forecast variance.

1 Introduction

Every centimeter of sea-level rise exposes roughly 1.6 million more people to annual coastal flooding (*Kulp and Strauss*, 2019). Without further adaptation, annual flood damages reach US$14–18 trillion per year at 0.9–1.1 m of rise (*Jevrejeva et al.*, 2018). Planning and adaptation for these risks depends on how skillfully we can predict the rate of sea-level rise. Forecasts published over four decades span a wide range of methods and outcomes, and the high-end estimates published since the Sixth Assessment Report of the Intergovernmental Panel on Climate Change (AR6) span 0.6–2.0 m of rise by 2100 (*Garner et al.*, 2026). Coastal planning leans on AR6 itself, which projects 0.4–0.7 m this century for low and middle emissions scenarios, but with deep uncertainty attributed to ice sheet processes not well represented in models (*Fox-Kemper et al.*, 2021; *Hirschfeld et al.*, 2023). The gaps in our understanding of ice sheet processes present significant challenges as continued warming

ensures that projections of sea-level rise will remain out of sample with respect to the modern observational record no matter the projection method. The central difficulty is therefore not building a model but calibrating it, testing it against data withheld from calibration, and reporting uncertainties in a form that separates the myriad sources. A recent community synthesis makes this case, calling for projections narrowed by observations and for uncertainty characterized by its structure and not only by its magnitude (*Felikson et al.*, 2025).

Sea-level projection models span a wide range of complexity, and their strengths are complementary. Process-based ice-sheet models solve the equations of ice flow, deformation, and fracture, and can test competing hypotheses about the mechanics of ice loss (*Schoof*, 2007; *Morlighem et al.*, 2024). Their parameters, however, are inferred by ill-posed inverse problems in which the data constrain only a few combinations of parameters (*Petra et al.*, 2014; *Isaac et al.*, 2015). These models have yet to reproduce prognostically the sixfold increase in ice-sheet mass imbalance observed over the past three decades (*Siegert et al.*, 2020), and many projections grow the Antarctic Ice Sheet over the satellite era, with outcomes sensitive to initialization (*Seroussi et al.*, 2020; *Fricker et al.*, 2025; *Goldberg et al.*, 2026).

At the other extreme, semi-empirical fits of the rate of global mean sea-level (GMSL) rise to global mean surface temperature (GMST) are transparent and anchored in the quantity being predicted, and they track the observed record at least as well as process-based projections made at the same time (*Rahmstorf*, 2007; *Vermeer and Rahmstorf*, 2009; *Rahmstorf et al.*, 2012; *Kopp et al.*, 2016; *Nerem et al.*, 2022). However, they give no insight into which processes drive the trends, and the fitted relationship need not hold when contributors respond on different timescales (*Church et al.*, 2013). Between these extremes sit component-level models, which constrain each physical contributor separately, trading spatial detail for parsimony and a direct link to data (*Perrette et al.*, 2013; *Kopp et al.*, 2014; *Mengel et al.*, 2016; *Nauels et al.*, 2017; *Wong et al.*, 2017; *Kopp et al.*, 2023).

No single class of model can be at once the most detailed in physical processes, the most transparent, and the most tightly constrained by data, so assuming any model type as authoritative discards the insights that a diverse set of models, modeling approaches, calibration methods, and independent lines of evidence provides. Ordered by complexity models can instead form a hierarchy, a practice long employed elsewhere in climate science, where ordering models from idealized to comprehensive has served understanding better than process-based models alone (*Held*, 2005; *Jeevanjee et al.*, 2017; *Maher et al.*, 2019). Each level in the hierarchy can be calibrated and tested against data before the next higher level of complexity is trusted, giving observations the primary role in refining projections that recent assessments of ice-sheet and sea-level projection have called for (e.g., *Felikson et al.*, 2025). *Aschwanden et al.* (2021) put the role of model diversity in explicit form, writing the predictive distribution as an integral of the projection over a distribution across model structures and then conditioning the result on observations. The integral presumes that the space being integrated over contains the true data-generating process, an assumption those authors themselves question, whereas a hierarchy requires only that models rely on assumptions different enough that their convergence or divergence is informative. Where structurally different models agree, a result rests on more than one set of assumptions, and agreement of that kind carries more weight than agreement among models that share their assumptions (*Lipscomb et al.*, 2025). Where they disagree, the divergence helps identify what

the data leave under-constrained and can be used to identify the observations that would provide further constraints. The simplest levels often carry the most transferable insight, because what they assume and include is small enough to state clearly, to test directly, and to be understood by those who use the model outputs (*Green and Armstrong*, 2015; *Makridakis et al.*, 2020). A hierarchy built this way complements and aids in the development of more complex, higher-fidelity process-based ice-sheet models rather than substituting for them.

Here we construct a data-driven, component-level forecast of 21st-century sea-level rise that sits at an intermediate level of model complexity between semi-empirical models and the complex process-based ice-sheet models. Where the observational record can constrain a component (e.g., mountain glacier mass loss), we define a parsimonious physical model calibrated by Bayesian inference against an independent record spanning decades. Greenland surface mass balance is projected with a statistical emulator of regional climate model simulations, and the remaining contributors, East Antarctic surface mass balance and terrestrial water storage, are adopted from published sensitivities and assessments. We call the framework SPECTRUM, for Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models. Unless otherwise noted, all sea-level values in the main text are global-mean 21st-century contributions referenced to the year 2000. Throughout, emission scenarios are labeled by their approximate median end-of-century global mean surface temperature rounded to the nearest integer and referenced to the 1850-1900 (pre-industrial) mean: 2 °C (SSP1-2.6), 3 °C (SSP2-4.5), and 4 °C (SSP3-7.0).

2 A data-driven, component-level projection framework

Each component of SPECTRUM is represented by its own model and is calibrated with independent datasets. Most components have an explicit or implicit GMST dependence that arises from the model calibration. The GMST dependence defines the forecast for each temperature-dependent component. Figure 1 summarizes the framework.

2.1 Component models, calibration, and validation

Decades of research have provided numerous data sets that allow the individual components to be calibrated. The supplemental materials provide a table of the data sets used in this study.

2.1.1 Greenland Ice Sheet

The Greenland Ice Sheet (GrIS) is the largest land-ice contributor and has lost mass every year since the 1990s (*Mouginot et al.*, 2019; *Mankoff et al.*, 2021). That loss is the imbalance between solid-ice discharge and surface mass balance (SMB), where SMB is the difference between mass gained by snow accumulation and mass lost to melting. GrIS discharge is sensitive to ocean temperature and surface melt runoff (*Straneo et al.*, 2013; *Slater et al.*, 2020; *Wood et al.*, 2021). Warm, subsurface waters thin and undercut the calving fronts of marine-terminating glaciers, driving increased calving and net terminus retreat. This effect is amplified where surface runoff drives mixing and buoyant upwelling along the calving face

(*Slater and Straneo*, 2022). Surface melt also lubricates glacier beds, but that effect is generally short-lived and localized as the subglacial hydrological system evolves to accommodate changes in meltwater flux (*Schoof*, 2010; *Tedstone et al.*, 2013; *Ultee et al.*, 2022).

Focusing on the strong influence of ocean-induced melting on discharge, we approximate the GrIS-wide discharge as

$$D = \gamma\, T_{\mathrm{ocean}}(t - \delta) + r_0, \tag{1}$$

where D is the GrIS discharge in sea-level equivalent units, γ is an effective discharge sensitivity to ocean temperature, δ is the lag between distal ocean warming and ice sheet response, r_0 is the background discharge rate at the baseline ocean temperature, and T_{ocean} is the ocean temperature anomaly in the ocean waters around Greenland relative to the 1995–2005 mean. Because D is the gross discharge flux rather than an anomaly, it becomes a contribution to GMSL only when combined with SMB.

We calibrate against total GrIS discharge, integrated across the 219 marine-terminating outlet glaciers resolved in the observations to trade spatial resolution for signal-to-noise in the data. Our goal is to capture ice-sheet scale discharge rather than the individual basins and glaciers. The calibration is carried out on the time-integrated form of Eq. 1, which gives the cumulative contribution of GrIS discharge to GMSL,

$$H_{\mathrm{D}}(t) = \int_{t_0}^{t} D\,dt', \tag{2}$$

with γ and r_0 set constant in time. Fitting the time-integrated (cumulative) record further improves the signal-to-noise ratio, because the cumulative signal grows linearly in time while the annual errors accumulate only as its square root. It also directly constrains the total mass lost, the quantity of interest in this study.

The calibration uses the 47-year ice discharge time-series of *Mouginot et al.* (2019) and ocean temperatures from EN4 (v4.2.2, g10 bias correction) (*Good et al.*, 2013), averaged over 200–500 m depth, the range most relevant to Greenland's tidewater grounding lines (*Slater et al.*, 2020), in the seas surrounding Greenland south of 80°N (58–80°N, 75–5°W). This geographic limit is set by the sparse subsurface sampling farther north. Because the data are resolved annually, the lag δ is in integer-year values. We fit (γ, r_0) by weighted least squares separately for a range of δ values, using the demeaned cumulative data (Fig. 2). Each fit carries a penalty on the mismatch between the modeled discharge rate at the end of the time-series and the rate observed over the final decade, which keeps the calibration consistent with the present-day discharge rate. The posterior is then formed as a Gaussian mixture of the fits for each δ, with weights derived from the Bayesian Information Criterion. This procedure places 99.5% of the weight on $\delta = 5$ yr and gives posterior medians $\gamma = 0.365$ mm SLE $\mathrm{yr}^{-1}\,{}^{\circ}\mathrm{C}^{-1}$ (0.312–0.419 mm SLE $\mathrm{yr}^{-1}\,{}^{\circ}\mathrm{C}^{-1}$) and $r_0 = 1.383$ mm SLE yr^{-1} (1.367–1.399 mm SLE yr^{-1}) (Fig. S1). The calibration target is a cumulative record, resulting in correlated errors, so we report parameter intervals from a covariance that accounts for that correlation and for the excess scatter of the residuals relative to the reported observational errors, such that reduced chi-square $\chi^2/\mathrm{dof} = 1.8$, where χ^2 is the error-weighted sum of squared residuals (Supplemental Materials). We validate the model against a discharge time-series spanning 1986–2023 that was withheld from calibration (*Mankoff et al.*, 2021) and the time-series from the Ice Sheet Mass Balance Intercomparison Exercise (IMBIE-3) spanning

1972–2023 (*Otosaka et al.*, 2026). The validation with *Mankoff et al.* (2021) is informative but weak because the *Mankoff et al.* (2021) time-series shares its underlying raw data with the *Mouginot et al.* (2019) calibration and differs from it only in processing. The IMBIE-3 validation is stronger. Its input-output estimates also share data with the calibration, but its altimetry and gravimetry estimates are independent of it. Our model falls within error of the validation data for discharge and total mass balance (Fig. 2). Because the SMB part of the model is fitted to the *Mouginot et al.* (2019) SMB, as described below, the total mass balance mainly tests the discharge model. The SMB emulator alone is tested against observed SMB in the Supplementary Material. The validation results suggest that our simple model, with two fitted parameters and a selected lag, captures the total contribution of GrIS discharge to GMSLR during the observational record and its sensitivity to ocean temperatures in Greenland's peripheral waters.

The fitted $\delta = 5$ yr is an effective lag for spatially averaged forcing and we therefore do not attempt to attribute it to a single process. It lies within the range of timescales spanned by the ocean-to-ice pathway, from exchange between the continental shelf and fjord interiors through the terminus response to undercutting and the inland transmission of retreat to the flux gates at which discharge is measured (*Straneo and Heimbach*, 2013; *Wood et al.*, 2021; *Robel et al.*, 2018). Because T_{ocean} is averaged over Greenland's peripheral waters, δ also absorbs the time required for a warming anomaly to become representative of conditions at outlet glaciers distributed around the ice sheet.

To our knowledge, no published study reports a Greenland-wide sensitivity of discharge rate to ocean temperature, so we derive a simple dimensional scaling to contextualize the inferred value $\gamma = 0.365\ \mathrm{mm\,SLE\,yr^{-1}\,{}^{\circ}C^{-1}}$. For a single outlet glacier i, the submarine melt rate scales approximately linearly with ocean thermal forcing, the ocean temperature above the in situ freezing point T_{ocean}, such that

$$\Delta \dot{m}_i = \kappa_i T_{\mathrm{ocean}}, \tag{3}$$

where κ_i is the melt sensitivity to thermal forcing averaged over the submerged calving face. Plume parameterizations calibrated against Greenland observations (*Xu et al.*, 2013; *Rignot et al.*, 2016; *Slater and Straneo*, 2022) give a maximum melt sensitivity near the grounding line of 90–260 $\mathrm{m\,yr^{-1}\,{}^{\circ}C^{-1}}$. The face-averaged rate is two to three times smaller (*Wood et al.*, 2021). We therefore take $\kappa = 36$–$104\ \mathrm{m\,yr^{-1}\,{}^{\circ}C^{-1}}$, with median $60\ \mathrm{m\,yr^{-1}\,{}^{\circ}C^{-1}}$. Submarine melt thins and undercuts the terminus, triggering calving and grounding-line retreat so that the total ice flux perturbation is a factor μ times the direct submarine melt flux. For Greenland's tidewater glaciers, observations and models suggest $\mu = 3$–10 (*Enderlin and Howat*, 2013; *Wood et al.*, 2021). The flux perturbation from a single glacier with submerged calving-front area A_f is then

$$\Delta \dot{Q}_i = \mu_i\, \rho_{ice}\, A_{f,i}\, \kappa_i T_{\mathrm{ocean}}, \tag{4}$$

where $\rho_{ice} = 900\ \mathrm{kg\,m^{-3}}$ is the mass density of ice. Summing over Greenland's $N = 219$ marine-terminating outlets (*Mouginot et al.*, 2019) we approximate the scale of γ as

$$\widetilde{\gamma} \sim \frac{1}{A_{\mathrm{ocean}}\, \rho_{fw} T_{\mathrm{ocean}}} \sum_{i=1}^{N} \Delta \dot{Q}_i, \tag{5}$$

where $A_{\text{ocean}} = 3.625 \times 10^{14}$ m^2 is the ocean area and $\rho_{fw} = 1000$ kg m^{-3} is the density of freshwater, which converts added ice mass to sea-level equivalent under the convention 1 mm SLE = 362.5 Gt used throughout this study. Submerged calving-front areas of 0.9–2.7 km^2 are reported for five West Greenland outlets (*Rignot et al.*, 2016). Evaluating Eq. 5 across the full parameter ranges gives

$$\widetilde{\gamma} = 0.05\text{–}1.5 \text{ mm SLE yr}^{-1} \,^{\circ}\text{C}^{-1}, \tag{6}$$

which contains our calibrated $\gamma = 0.365$ mm SLE yr^{-1} °C^{-1}. We report Eq. 6 as a rough order-of-magnitude check on the inferred value intended for intuition building. The range spans a factor of 30 because it is the product of four quantities that are not measured in aggregate, and the factorization $\sum_i \mu_i A_{f,i} \kappa_i \approx N \, \bar{\mu} \, \bar{A}_f \, \bar{\kappa}$ assumes these are uncorrelated across outlets, whereas the deepest and widest fronts occupy the warmest fjords. Finally, the calibrated γ is an effective sensitivity that absorbs the covarying atmospheric pathway because increased subglacial runoff amplifies plume-driven melting alongside ocean warming.

2.1.2 Thermosteric

The ocean is the dominant reservoir for the excess heat accumulated by the climate system (*Cheng et al.*, 2017), and the resulting expansion – the thermosteric contribution – is currently the largest single contributor to observed sea level rise (*Frederikse et al.*, 2020). Both heat exchange between ocean and atmosphere and ocean dynamics set the structure of a simple model of the thermosteric effect. Put simply, the ocean warms with a delay relative to the atmosphere, heat enters at the sea surface and is carried downward by turbulent mixing and by the overturning circulation, causing temperature of the interior to reflect the history of surface warming. Downward transfer spans a wide range of timescales. The upper few hundred meters exchange heat with the atmosphere over years to decades, whereas water below the thermocline is renewed over centuries. We therefore represent the ocean as two layers, a fast upper layer coupled to the atmosphere and a slow deep layer coupled only to the upper layer, following the structure of *Geoffroy et al.* (2013). This structure reproduces both the lag between surface warming and expansion and the continued expansion that follows a stabilization of surface temperature, where the deep layer draws heat from the upper layer after the upper layer has equilibrated.

Let us first consider an upper layer of thickness $h_u = 700$ m warmed by $\Delta\theta(z,t)$, which raises steric height of the ocean over time as

$$H_{\text{u}} = \int_0^{h_u} \alpha_{sw} \Delta\theta \, dz, \tag{7}$$

where $\alpha_{sw}(z)$ is the thermal expansion coefficient of seawater, which is depth dependent due to its temperature dependence. The temperature anomaly profile is defined as $\Delta\theta = S_u \varphi$, where $S_u(t)$ is the amplitude of the anomaly and $\varphi(z)$ is a dimensionless shape factor. Expanding $\alpha_{sw}(z) = \alpha_{0_{sw}}(z) + \alpha'_{sw}\Delta\theta$, with $\alpha'_{sw} = \partial\alpha_{sw}/\partial\theta$, we rewrite Eq. 7 as

$$H_{\text{u}} = a_u S_u^2 + b_u S_u, \tag{8}$$

where $a_u = \int_0^{h_u} \alpha'_{sw} \varphi^2 dz$ and $b_u = \int_0^{h_u} \alpha_{0_{sw}} \varphi dz$. The linear term b_u is the expansion of the upper layer for a given warming while the quadratic term a_u comes from the temperature dependence of α_{sw}. As GMST, $T(t)$, changes, the amplitude evolves as

$$\frac{dS_u}{dt} = \frac{T - S_u}{\tau_u}, \tag{9}$$

where τ_u is the characteristic time over which the thermal state S_u relaxes to GMST. Equation 9 shows that S_u is an average of past temperatures, weighted toward the recent past over a time window set by τ_u. The upper layer therefore smooths year-to-year fluctuations in GMST and follows longer trends with a delay comparable to τ_u.

The deep layer follows the same construction, with $\Delta\theta_d(z,t) = S_d(t)\,\varphi_d(z)$ over 700–2000 m depth and $b_d = \int_{h_u}^{h_d} \alpha_{0_{sw}}\,\varphi_d\,dz$, with $h_d = 2000$ m. We drop the quadratic term because S_d remains small enough that the temperature dependence of the thermal expansion coefficient can be neglected. Warming from GMST reaches the deep layer only after passing through the upper layer, so the thermal state S_d relaxes toward S_u as

$$\frac{dS_d}{dt} = \frac{S_u - S_d}{\tau_d}, \tag{10}$$

where we fixed $\tau_d = 150$ yr, the CMIP5 multi-model mean deep-ocean adjustment time (*Geoffroy et al.*, 2013, Table 4), because the 21-year 0–2000 m data record is too short to constrain a centennial relaxation time. Adding this layer gives the total steric response for the upper 2000 m as,

$$H_{ud}(t) = H_{\mathrm{u}}(t) + b_d\,S_d + \epsilon_u(t - t_0), \tag{11}$$

which we fit jointly to the 0–700 m and 0–2000 m records. We tacked on $\epsilon_u(t - t_0)$ to absorb instrument drift during calibration. Finally, to account for expansion below 2000 m depth, we define

$$H_{\mathrm{full}}(t) = H_{ud}(t) + \dot{H}_b(t - t_{\mathrm{ref}}), \tag{12}$$

where $\dot{H}_b = 0.07 \pm 0.04\,\mathrm{mm\,yr^{-1}}$ (*Purkey and Johnson*, 2010) is a constant rate for depths below 2000 m, referenced to year $t_{\mathrm{ref}} = 2000$. This layer is not observed at annual resolution, so its rate is adopted rather than calibrated.

We fit the two-layer model in Eq. 11 jointly to the thermosteric sea level estimates at 0–700 m (1955–2025) and 0–2000 m (2005–2025) (*Levitus et al.*, 2012). We use Bayesian inference (*Tarantola*, 2005) in which MCMC samples parameter combinations $\boldsymbol{\phi} = (a, b_u, b_d, \epsilon_u, \tau_u, \sigma_{\mathrm{extra}})$ with probability density proportional to the posterior (Fig. S2 of the Supplementary Materials). Assuming the residual at each point is normally distributed about the model with variance σ_i^2, the negative log-likelihood takes the form $\frac{1}{2}\sum_i [(r_i(\boldsymbol{\phi})/\sigma_i)^2 + \ln \sigma_i^2]$. Adding a negative log-prior for τ_u, whose log-normal prior (defined below) has mean μ_{τ_u} and standard deviation s_{τ_u} in $\ln \tau_u$, gives the objective function

$$\mathcal{J}(\boldsymbol{\phi}) = \sum_i \left[\left(\frac{r_i(\boldsymbol{\phi})}{\sigma_i} \right)^2 + \ln \sigma_i^2 \right] + \left(\frac{\ln \tau_u - \mu_{\tau_u}}{s_{\tau_u}} \right)^2, \tag{13}$$

which is twice the negative log-posterior up to an additive constant. MCMC visits parameter combinations $\boldsymbol{\phi}$ with probability proportional to $\exp[-\mathcal{J}(\boldsymbol{\phi})/2]$. Here, $r_i(\boldsymbol{\phi})$ is the residual

between the demeaned model and demeaned observation at point i, and σ_i^2 is that point's total variance. The point variance is defined by the quadrature sum of the formal measurement error, a structural uncertainty retained after a bias correction in the NOAA thermosteric observations (*Li et al.*, 2022), and σ_{extra}, a noise term, shared across both the 0–700 m and 0–2000 m records, that absorbs misfit beyond the reported observational error. The MCMC sampling uses package emcee with 64 walkers, 6000 burn-in steps, and 30,000 production steps to ensure convergence. The relaxation time of the upper layer, τ_u, has an informative prior, $\log(\tau_u) \sim \mathcal{N}(\ln 8, 0.5^2)$, or $\mu_{\tau_u} = \ln(8\text{ yr})$ and $s_{\tau_u} = 0.5$ yr with $\mathcal{N}$ indicating a normal distribution, consistent with the range in *Geoffroy et al.* (2013). The model parameters b_u and τ_u are correlated in the likelihood—a slower upper-layer response can be partly offset by a larger sensitivity to fit the same trajectory—but this partial degeneracy is managed by the prior on τ_u. Five parameters — a, b_u, b_d, ϵ_u, and $\ln(\sigma_{\text{extra}})$ — get uniform priors with wide bounds (Fig. S2). The posterior median and 90% credible intervals are: $a = 20$ [10.1, 31.2] mm °C^{-2}, $b_u = 51$ [40.6, 75.4] mm °C^{-1}, $b_d = 68.6$ [37.0, 104.3] mm °C^{-1}, $\epsilon_u = 0.023$ [−0.112, 0.140] mm yr^{-1}, $\tau_u = 11.6$ [6.2, 25.3] yr, and $\sigma_{\text{extra}} = 2$ [1.7, 2.3] mm (Fig. S2). The calibrated model reproduces both records with RMS residuals of 2.3 mm for depths 0–700 m and 1.3 mm for 0–2000 m, against 42 and 30 mm of observed steric rise over the respective records (Fig. 3).

We validate the full-depth model in Eq. 12 against two independent global-mean steric sea level records, both withheld from calibration (Fig. 3c,f). *Frederikse et al.* (2020) reconstruct steric sea level for 1900–2018. Over 1957–2018 they draw, for each ensemble member, from three gridded products built on the same archive of in situ temperature and salinity profiles (*Levitus et al.*, 2012; *Ishii et al.*, 2017; *Cheng and Zhu*, 2016). The first of these is the data our model is calibrated on, so over this period the comparison measures consistency between separate analyses of largely shared data. Before 1957, when in situ coverage is too sparse to resolve global steric change, the comparison is instead to their pre-1957 segment, which is built on the *Zanna et al.* (2019) reconstruction of ocean heat storage, inferred from sea-surface temperature anomalies and heat-transport pathways from an ocean reanalysis. *Zanna et al.* (2019) also supplies their expansion below 2000 m over the full 1900–2018 period, where we adopt the constant rate of *Purkey and Johnson* (2010). The pre-1957 segment is therefore the part of this comparison that is independent of the data used in calibration. Our model agrees with the reconstruction to an RMS difference of 2.6 mm over 1957–2018 and 3.8 mm before 1957. Our second validation comes from *Dangendorf et al.* (2024) who assimilate 516 tide-gauge records and satellite altimetry in an ensemble Kalman smoother that solves for GMSL over 1900–2021. To estimate the thermosteric contribution, they subtract barystatic, glacial isostatic adjustment, and inverse-barometer contributions from GMSL. No temperature or salinity measurement enters this calculation, so the comparison tests our model against a steric estimate that shares none of the in situ ocean data used to calibrate it. Our model agrees with their reconstruction to an RMS difference of 3.5 mm over 1957–2021 and 5.0 mm before 1957 (Fig. 3c,f).

2.1.3 Glaciers

The rate of glacier mass loss has been shown to be approximately linear with warming (*Rounce et al.*, 2023; *The GlaMBIE Team et al.*, 2025). In the interest of developing consis-

tent uncertainty quantification for this study, we perform our own Bayesian inference of the cumulative glacier sea-level contribution using the Glacier Mass Balance Intercomparison Exercise (GlaMBIE) time-series from 2000–2023. In level space, the model is

$$H_{\mathrm{gl}}(t) = b_{gl} \int_{t_0}^{t} T d\tau + c_{gl}(t - t_0) + H_{gl_0} + \varepsilon_{gl}(t), \tag{14}$$

where T is the GMST anomaly, evaluated at the annual resolution of the GlaMBIE calendar-year-mean, $t_0 = 2000$ is the study's baseline year when H_{gl_0} is the model's level, and $\varepsilon_{gl}(t)$ is jointly distributed as $\mathcal{N}(0, \Sigma_{gl})$, with Σ_{gl} the true covariance matrix of the cumulative data, built from the GlaMBIE annual uncertainties. We apply a weakly informative prior $b_{gl} \sim \mathcal{N}(0.61, 2.0)$ mm yr^{-1} °C^{-1}, truncated at $b_{gl} \geq 0$. The prior mean for b_{gl} is drawn from *Rounce et al.* (2023), whose reported end-of-century glacier contributions of 0.7 to 2.23 mm yr^{-1} across warming scenarios of 1.5 to 4 °C imply a sensitivity near 0.61 mm yr^{-1} °C^{-1}. The standard deviation is wide enough that the prior is only weakly informative, and the non-negativity truncation recognizes that glaciers lose mass as the climate warms. We constrain the fit for c_{gl}. Glaciers entered the industrial era close to balance with the slowly cooling climate of the preceding centuries (*Roe et al.*, 2021), so we set the rate to zero at a balance temperature $T_{gl,eq}$, which gives $c_{gl} = -b_{gl} T_{gl,eq}$. The model we fit is therefore

$$H_{\mathrm{gl}}(t) = b_{gl} \int_{t_0}^{t} (T - T_{gl,eq})\, d\tau + H_{gl_0} + \varepsilon_{gl}(t). \tag{15}$$

The prior on $T_{gl,eq}$ is normal, centered on the 1850–1879 mean GMST of −0.86 °C relative to 1995–2005, with a standard deviation of 0.15 °C. With c_{gl} free, the fitted model gains glacier mass before about 1935. For the MCMC, we use package emcee with 32 walkers, 2000 burn-in steps, and 4000 production steps. The posterior median sensitivity is $b_{gl} = 0.72$ [0.58, 0.94] mm yr^{-1} °C^{-1}, the balance temperature is $T_{gl,eq} = -0.81$ [−1.05, −0.58] °C, and the committed rate is $c_{gl} = 0.59$ [0.52, 0.65] mm yr^{-1}, intervals that the balance prior narrows (Fig. 4a). We validate the glacier model against the global glacier mass changes of *Zemp et al.* (2019) for 1962–1999, which are derived from glaciological and geodetic observations and are not used in the calibration. Over these 38 years the model gives a mean rate of 0.31 [0.19, 0.40] mm yr^{-1}, compared with an observed 0.32 [−0.02, 0.66] mm yr^{-1} (Fig. 4a). We compare only the years before 2000 because over 2000–2016, where both records exist, the loss rate of *Zemp et al.* (2019) exceeds that of GlaMBIE by 0.18 mm yr^{-1}.

2.1.4 Antarctic Peninsula

The Antarctic Peninsula component uses the same level-space Bayesian framework as glaciers, fit at annual resolution to the IMBIE-3 Peninsula time-series spanning 1979–2023 (45 annual points; *Otosaka et al.*, 2026), with the rate–temperature model $\dot{H}_{\mathrm{p}} = a_p T^2 + b_p T + c_p$. The Peninsula contribution in this work reflects combined SMB and glacier discharge responses to warming. As with glaciers, H_{p_0} is anchored at the record's own first point (1979), and the true covariance of this cumulative record, built from the IMBIE-3 uncertainties, is used in the likelihood. Unlike glaciers, we also fit a model-inadequacy term $\sigma_{p,\mathrm{extra}}$ to account for annual scatter beyond the IMBIE-3 reported uncertainty. The emcee implementation uses

32 walkers, 2000 burn-in steps, and 4000 production steps. We fit both the linear ($a_p = 0$) and quadratic model with priors $b_p \sim \mathcal{N}(0,\, 2.0)$ mm yr^{-1} °C^{-1} and $c_p \sim \mathcal{N}(0.3, 1)$ mm yr^{-1}. The Bayesian Information Criterion favors the linear model with posterior median sensitivity $b_p = 0.044$ mm yr^{-1} °C^{-1} [−0.022, 0.106] and committed rate $c_p = 0.056$ mm yr^{-1} [0.039, 0.072] (Fig. 4b). The confidence interval spans zero, so these data do not, on their own, resolve the sign of the temperature sensitivity.

2.1.5 East Antarctica

The EAIS component follows the same corrected level-space framework as glaciers and the Peninsula, fit at annual resolution to the IMBIE-3 EAIS record (1979–2023, 45 annual points; (*Otosaka et al.*, 2026)), with H_{e_0} anchored at the record's own first point and the true covariance of the cumulative record, built from the IMBIE-3 uncertainties. The linear sensitivity b_e is given a symmetric prior, $b_e \sim \mathcal{N}(-0.17,\, 0.5)$ mm yr^{-1} °C^{-1}. The prior mean is drawn from an independent literature surface-mass-balance sensitivity ($C_T = 60 \pm 20$ Gt yr^{-1} °C^{-1}; *Frieler et al.*, 2015; *Ligtenberg et al.*, 2013). The posterior sensitivity is $b_e = -0.016$ mm yr^{-1} °C^{-1} [−0.15, 0.11] and committed rate $c_e = 0.007$ mm yr^{-1} [−0.03, 0.04] (Fig. 4c).

2.1.6 West Antarctica

Unlike glaciers, the Antarctic Peninsula, and EAIS, we do not fit a rate–temperature regression to WAIS over the observational record because heat in the oceans has been the major driver of WAIS mass loss, and oceanographic data sets are too sparse to allow for a reliable ocean-driven fit. We instead fit a quadratic in time to the 1979–2023 IMBIE-3 mass balance time series (*Otosaka et al.*, 2026). The acceleration a is given a symmetric prior, $a \sim \mathcal{N}(0,\, \sigma = 0.05$ mm yr$^{-2})$. The quadratic yields an acceleration of 0.0140 mm yr^{-2} [0.0115–0.0165] and a rate in year 2000 of 0.226 mm yr^{-1} [0.200–0.252] (Fig. 4d). Snowfall anomalies in the Amundsen Sea Embayment modulate this record over a few years at a time (*Davison et al.*, 2023). Removing the two anomalies they report (2009–2013 and 2019–2020) lowers the extrapolated contribution at 2100 from 93 to 83 mm, well within the S1 uncertainty described below, so we fit the record as reported. We set the rate of WAIS loss to zero prior to 1979, when the IMBIE-3 time series begins and the fit to the data shows no mass loss (a rate of −0.06 mm yr^{-1}).

2.2 Forecasting methods and validation

The forecasts presented here use the IPCC AR6 GMST trajectories for different emission scenarios, though they can be applied to any GMST trajectory. We focus on the policy-relevant emission scenarios, which are labeled by their approximate median end-of-century global mean surface temperature anomaly (relative to preindustrial) rounded to the nearest integer: 2 °C (SSP1-2.6), 3 °C (SSP2-4.5), and 4 °C (SSP3-7.0). We initialize the GMSL forecast using the GMSL level, rate, and acceleration observed in the satellite-era and blend in the component-model sum in rate space, such that

$$\dot{H}_{\text{forecast}} = w\dot{H}_{\text{obs}} + \big[1 - w\big]\dot{H}_{\text{comp}}, \tag{16}$$

where $\dot{H}_{\rm obs}$ is the rate from the satellite-era quadratic fit of *Hamlington et al.* (2024); *Nerem et al.* (2022) and $\dot{H}_{\rm comp}$ is the rate of the sum of the calibrated components. The blending weight $w(t)$ is a sigmoid

$$w = 1 - \Theta\left(\frac{t - t_r}{\tau_r}\right), \tag{17}$$

where $\Theta(x) = (1 + e^{-x})^{-1}$ is the standard logistic function, $t_r = 2035$, and $\tau_r = 5$ yr. At $t \ll t_r$, $w \approx 1$ and the forecast follows the observed quadratic trajectory. At $t \gg t_r$, $w \approx 0$ and the forecast follows the component models. The transition spans approximately 22 years, meaning that 10%–90% of the weight shift occurs over $\pm 2.2\tau_r \approx \pm 11$ yr about t_r. We chose $t_r = 2035$ because it is more than a quarter and less than a half of the GMSL satellite altimetry record into the future, a period when the extrapolation of observations should have strong forecasting skill. The tapering recognizes the expected reduction in forecasting skill of the simple data extrapolation over timescales longer than roughly half of the observational window. Pushing t_r to 5 years later (approximately half of the calibration window and arguably the latest viable choice) produces forecasts that are indistinguishable within uncertainty, suggesting the results are largely insensitive to the specific value of t_r within the range of 10-15 years into the future.

The GMSL forecast is obtained by integrating Eq. (16) forward from the observed level at the forecast origin ($t_0 = 2025$, the most recent NASA altimetry data as of this writing), so that

$$H_{\rm forecast}(t) = H_{\rm obs}(t_0) + \int_{t_0}^{t} \dot{H}_{\rm forecast}(t')\, dt'. \tag{18}$$

The blending is done sample-by-sample across the 2000-member Monte Carlo ensemble, preserving correlations between the quadratic-fit parameter uncertainty and the component-model uncertainty. Uncertainty in $\dot{H}_{\rm comp}$ is propagated by Monte Carlo summation of the components. Each component has its own 2000-member ensemble of level-space projections $H_k(t)$, with parameters drawn independently from that component's posterior or, for components using prescribed literature sensitivities rather than data calibration, from the corresponding parameter distributions. Member i of every component is driven by the same GMST trajectory, a single percentile path through the IPCC AR6 5–95% range held for the whole century, with zero offset through 2024 when warming is observed, and the same percentile is used in every SSP. The total level is formed sample-by-sample, $H_{{\rm comp},i}(t) = \sum_k H_{k,i}(t)$ for $i = 1, \ldots, 2000$, so parameter draws are independent across components while the within-scenario warming path is shared. The component rate $\dot{H}_{\rm comp}$ used in Eq. (16) is obtained by finite-differencing this summed ensemble in time before blending with the observational quadratic rate.

2.2.1 Component-level GMST dependence

Some of the SPECTRUM model components described in Section 2.1 lack explicit dependences on GMST. These components include the Greenland glacier discharge, which depends on the temperature of the ocean surrounding Greenland, and the SMB of Greenland and EAIS.

To project GrIS discharge model forward in time, we calibrate a transfer function between the Greenland regional surface temperature anomaly and the ocean temperature and then

apply a normal distribution of Arctic amplification factors to calculate the Greenland regional surface temperature from projected GMST under different emission scenarios. The linear transfer function on annual-mean observations is

$$T_{\text{ocean}}(t) = \xi T^{\text{Gr}}_{\text{surface}}(t) + \chi + \varepsilon_{go}(t), \tag{19}$$

where $T^{\text{Gr}}_{\text{surface}}$ is the Greenland regional surface temperature anomaly and ε_{go} is a zero-mean residual. Both series are anomalies relative to the 1995–2005 baseline. An OLS fit over 52 common annual values gives $\xi = 0.148 \pm 0.027$, $\chi = -0.051 \pm 0.031$°C, and standard deviation $\sigma_{\varepsilon_{go}} = 0.20$°C of the residual ε_{go}.

For projections, SSP GMST trajectories are converted to Greenland regional surface temperature using an Arctic amplification factor $AA = 2.5 \pm 0.5$. The Arctic as a whole warmed nearly four times faster than the globe over 1979–2021 (*Rantanen et al.*, 2022), but the ratio of Greenland regional to global warming in the observations we use is lower and depends on the averaging window, spanning 2.1 to 2.6 for windows that start between 1970 and 1990. Greenland regional surface temperatures are used in Eq. (19) to obtain projected T_{ocean}, which plugs into the the delayed integral in Eq. (2). Each of the 2,000 Monte Carlo samples draws AA $\sim \mathcal{N}(2.5, 0.5^2)$, $\xi \sim \mathcal{N}(0.148, 0.027^2)$, and $\chi \sim \mathcal{N}(-0.051, 0.031^2)$ jointly with (γ, r_0) and δ from the discharge posterior, so that Arctic amplification, transfer-function, and discharge parameter uncertainties all propagate into the projections. A single draw of (ξ, χ, γ, r_0) is applied to all SSP scenarios, so that inter-SSP differences isolate the effect of the emissions pathway from parameter uncertainty.

For Greenland SMB, the forecast uses a statistical emulator of the regional climate model MARv3.12 driven by seven CMIP6 global climate models (*Boberg et al.*, 2026). For each global model, we fit the annual SMB contribution as a quadratic using the modeled GMST anomaly T relative to 1995–2005, $\dot{H}_{\text{smb}} = a_{gris}T^2 + b_{gris}T + c_{gris}$, pooling its historical and SSP simulations (Supplementary Material, Figs. S6–S8). Each Monte Carlo member draws one global model, with equal weight, and one posterior draw of a_{gris} and b_{gris}. Driven by observed GMST, the emulator does not reproduce a slowly varying departure of the observed SMB from its response to warming. Relative to 1995–2005, the *Mouginot et al.* (2019) SMB lies about 81 Gt yr^{-1} above the emulator over 1972–1985 and 106 Gt yr^{-1} below it over 2010–2018 (Supplementary Material). We represent this departure as a drift state $\delta(t)$ that relaxes toward zero over a timescale τ, so that $\dot{H}_{\text{smb}} = a_{gris}T^2 + b_{gris}T + c_{gris} + \delta(t) + \varepsilon(t)$, where ε is year-to-year weather. For each member, c_{gris} and the path of δ are estimated with a Kalman filter from the *Mouginot et al.* (2019) SMB over 1972–2018. The 47-year record does not constrain τ. We therefore set it with a lognormal prior whose 90% range is 5 to 25 yr, which allows about two relaxation times within the record, and obtain $\tau = 12$ [6, 26] yr. In 2018 the state places the SMB 27 Gt yr^{-1} below the emulator, with a 90% range from 167 Gt yr^{-1} below to 108 Gt yr^{-1} above, and it relaxes toward zero thereafter. The drift state does not change the emulator's response to warming. We also include the SMB–elevation feedback, an additional loss of 8 ± 5% at the reference simulation of *Fettweis et al.* (2013), scaled by each member's cumulative SMB anomaly since 2000. Because MAR is the only regional climate model in the training set, the spread between regional climate models is not sampled (*Glaude et al.*, 2024). When each global model is left out in turn, the other six predict its 2015–2100 SMB contribution within the 90% predictive range for all 13 withheld simulations.

For EAIS SMB, we use a linear model ($C_{T^2} = 0$) and the literature-derived accumulation sensitivity $C_T = 60 \pm 20$ Gt yr^{-1} °C^{-1} of GMST (*Frieler et al.*, 2015; *Ligtenberg et al.*, 2013), reflecting the Clausius–Clapeyron scaling of snowfall with warming ($C_T > 0$ corresponds to EAIS mass gain through accumulation). Data from IMBIE-3 show EAIS in approximate balance (Fig. 4c), so we set $\dot{M}_0 = 0$. The EAIS contribution is small (-14 to -30 mm at 2100, depending on scenario) relative to the total, meaning the GMSL projection results of this study are largely insensitive to these parameter choices.

Terrestrial water storage projections are adopted directly from the IPCC AR6 rather than independently modeled, as the processes governing groundwater depletion, reservoir impoundment, and land hydrology changes are outside the scope of this framework and the data are insufficient to allow for a data-driven approach as with other components. At each decadal interval in the IPCC projections, we fit a Gaussian distribution to the reported median and inter-quantile spread and draw 2,000 Monte Carlo samples, which are then linearly interpolated to annual resolution. For the hindcast and budget-closure diagnostics we use the *Frederikse et al.* (2020) TWS estimate with Monte Carlo sampling from their reported uncertainties, but only from 2003 onward. Their TWS is derived from GRACE and GRACE-FO observations over 2003–2018, partitioned from the total land-mass signal. Before 2003 it is a composite of a reconstruction of climate-driven water storage and process models for reservoir impoundment and groundwater depletion rather than a measurement. We therefore set TWS to zero before 2003, rebased so the cumulative contribution is continuous across the transition, and we note that the pre-2003 component sum consequently omits a real contribution to sea level. TWS varies little among scenarios and its contribution is small relative to the total, so it does not materially affect the uncertainty budget.

2.2.2 West Antarctic Ice Sheet

West Antarctica dominates the uncertainty in sea-level projections because its major outlet glaciers, particularly Thwaites and Pine Island, lie close to a threshold at which small changes in ice geometry could trigger self-sustaining retreat (*Joughin et al.*, 2014; *Pattyn and Morlighem*, 2020; *Fricker et al.*, 2025). Whether that threshold has been, or will soon be, crossed depends on glacier-specific details that remain only partly observed. We do not attempt to constrain a GMST sensitivity for WAIS. Such an effort is beyond the scope of this work. Instead, we develop a two-scenario mixture model to represent a range of potential GMSL contributions from WAIS in the 21st century. Based on growing evidence that WAIS mass loss is primarily sensitive to internal ice dynamics and deep ocean temperatures that are likely to change slowly in the coming decades, we make our WAIS projections independent of the warming trajectory. Thus, the range of 21st-century outcomes is controlled by ice dynamics that are insensitive to emissions over our forecasting horizon in the SPECTRUM model, consistent with previous studies (*Naughten et al.*, 2023; *Coulon et al.*, 2024; *Rosier et al.*, 2025). This independence is limited to the 21st century, because warming over this century largely sets the Antarctic contribution in the centuries that follow (*Lowry et al.*, 2021; *Seroussi et al.*, 2024). Some studies find emissions dependence even before 2100, with major WAIS retreat beginning this century in some simulations under 3 °C of warming but not under 2 °C (*DeConto et al.*, 2021). A recent emulation of the ISMIP6-2300 Antarctic ensemble, calibrated against the observed 2002–2021 Antarctic mass-loss rate, finds that higher

emissions raise the 2100 Antarctic contribution with probability 0.89 (*Lin et al.*, 2026). The magnitude of that dependence is small over the pathways we consider. Their drift-corrected posterior median for the whole ice sheet rises by 5 mm from SSP1-1.9 to SSP2-4.5, a shift small compared with our S1 90% credible interval of 93 mm [−3, 191]. The sensitivity is strongly nonlinear, 1.2 mm $°C^{-1}$ below 2 °C of warming since 2015, or about 3.3 °C above pre-industrial, and 17.9 mm $°C^{-1}$ above it. Their emulator predicts the Antarctic total rather than West Antarctica alone, so part of that dependence arises from the surface mass balance response that we carry separately for EAIS.

In Scenario 1 (S1), mass loss continues along the observed trajectory (Fig. 4d). Extrapolating the trend gives a median GMSL contribution of 93 mm by 2100, 8 mm above the median of the ISMIP6 emulator, 85 mm (*Edwards et al.*, 2021). The ISMIP6 emulator is the only model in AR6 with a symmetric distribution. This median falls between the modes of the LARMIP-2 assessment (*Levermann et al.*, 2020) — with mode 73 mm, median 137 mm, and skewness 2.42 — and the structured expert judgment (*Bamber et al.*, 2019), with mode 120 mm, median 207 mm, and skewness 2.40. A 45-year observational record cannot itself constrain how much additional error should accumulate from extrapolating a further 76 years beyond it, so a confidence interval built from the fitted parameters alone would understate the true uncertainty in this extrapolation. We instead treat the reported growth in uncertainty beyond the last observation in the ISMIP6 emulator as an external estimate of the extrapolation error, applying it as a prediction error *Tarantola* (2005), and add it to our parameter uncertainty. This widens our 90% CI at 2100 to [−3, 191] mm, closely matching the range of the ISMIP6 emulator of [−11, 182] mm. The S1 probability distribution, exceedance curve, and time-series are shown in Fig 5 by the curve with the probability of S1, p(S1), at 100%.

Scenario 2 (S2) is intended to represent a future where mass loss can accelerate well beyond the current observed rates. We define the distribution in year 2100 with a lower bound (5th percentile) of 93 mm, equal to the S1 median. The S2 upper bound (95th percentile) is 1.3 m, approximately the IPCC AR6 low-confidence Antarctic Ice Sheet 95th percentile under SSP5-8.5 in the year 2100 (*Fox-Kemper et al.*, 2021). The distribution is right-skewed, and defined as skew-normal in log-space with shape parameter $\lambda = 3$, reflecting the expectation that instability skews outcomes toward greater loss (*Robel et al.*, 2019). A sensitivity analysis indicates that the results we present are largely insensitive to the chosen value of λ within the expected reliability of the distributions. The S2 probability distribution, exceedance curve, and time-series are shown in Fig 5 as the p(S1) = 0 curve.

Emerging research on physical processes not yet represented in ice-sheet models supports the possibility of loss well above the observed trajectory. Examples include seawater intrusion beneath grounded ice (*Rignot et al.*, 2024; *Bradley and Hewitt*, 2024) and the evolution of subglacial water pressure (*Zhao et al.*, 2025). The bound implied by the ISMIP6 ensemble itself is far lower. *Lin et al.* (2026) identify a cascade of ice-sheet model assumptions that remains consistent with the observed mass-loss rate and yields 157 mm [98, 254] for the whole ice sheet at 2100 under SSP5-8.5, roughly a fifth of our S2 95th percentile for WAIS alone. That ensemble excludes marine ice cliff instability, seawater intrusion, subglacial hydrology, and ice damage, so the difference measures the weight our S2 places on processes absent from current ice-sheet models rather than a disagreement about those models. The S2 95th percentile adopted here should therefore not be read as an upper bound on plausible loss,

and Figure 6 shows how the WAIS contribution changes with it.

The final probability density is a simple mixture of S1 and S2. Its median, 95th percentile, and other quantiles are determined primarily by p(S1) and the upper bound (95th percentile) of S2. These two parameters are closely related in a simple mapping to the median and 95th percentile of the mixture (Fig. 6), which allows a single control variable for most use cases (Fig 5). The trajectory of WAIS mass loss through time is set by a power law in time that starts at the observed rate and accelerates toward the 2100 value, with median exponent $\beta = 2$, chosen so the S2 95th percentile stays within the IPCC AR6 medium-confidence range for the Antarctic Ice Sheet until about 2050 and exceeds it afterward. We chose p(S1) as the control variable and present two cases at the bounds of the IPCC calibrated likelihood scale: a 'slow WAIS' case with p(S1) = 90%, in which continued mass loss along the observed trajectory is very likely, and a 'fast WAIS' case with p(S1) = 10%, in which it is very unlikely. The 10% weight retained on the accelerating branch in the slow-WAIS case reflects that the observational record cannot fully retire the possibility of rapid mass loss.

3 Results

Our results contain both validation and projections. Hindcasting skill should be a weighting factor in considering the validity of model's projections as it provides a means to test forecasting skill against data and provides insight into the forecasting horizon over which the model results can be trusted. As a result, we present our validation attempts before the projections.

3.1 Validation

We take a multifaceted approach to model validation by considering budget closure and hindcasts against independent data sets.

3.1.1 Total GMSL budget closure

Satellite altimetry data collected from 1993–2025 provide observations of GMSL (*Beckley et al.*, 2017; *Hamlington et al.*, 2020) that can be compared with the sum of the independently calibrated model components (Fig. 7). Total GMSL is withheld from calibration to serve as a validation data set. For Greenland SMB we use the RACMO-derived SMB of *Mouginot et al.* (2019) through 2018 and the emulator thereafter. Terrestrial water storage is adopted from *Frederikse et al.* (2020) from 2003 onward, the period over which their estimate derives from GRACE observations, and set to zero before. The uncharted-glacier term described in the next section is zero from 2000 onward.

The component sum closes 93–101% of the observed GMSLR rate over three epochs (1993–2025, 2000–2025, 2010–2025), offering validation that the individually calibrated components together approximate GMSL during the satellite altimetry record (Fig. 7). Because each component is calibrated independently, we treat their rate uncertainties as independent and combine them in quadrature. Each component's individual uncertainty comes directly

from its Monte Carlo ensemble. For a given epoch, we fit a linear trend to every ensemble member and take the standard deviation of the resulting trends. Repeating this for each epoch lets the combined uncertainty differ from one epoch to the next, rather than using one fixed value throughout the satellite record. By this measure, the residual is 0.93σ, 0.35σ, and 1.61σ of the combined component uncertainty for 1993–2025, 2000–2025, and 2010–2025. The largest shortfall, over 2010–2025, comes mostly from Greenland. After the observed SMB record ends in 2018, the modeled SMB is higher than observed, so the model understates Greenland's recent losses.

3.1.2 Hindcasting the level, rate, and transient sensitivity

We validate the combined GMST-sensitive model by two hindcasting approaches. Both are generated by forcing the model components with observed GMST back to 1900 (*Rohde and Hausfather*, 2020), with an emphasis on post-1950 when increasingly reliable GMSL data are available. The glacier hindcast also includes glaciers that are missing from the inventories on which the GlaMBIE record is built, because they are too small to map or because they melted away after 1901. *Parkes and Marzeion* (2018) estimate that these uncharted glaciers contributed 0.17 to 0.53 mm yr^{-1} over 1901–1990, most of it early in the century, and 0.08 to 0.21 mm yr^{-1} over 1993–2010. We add this contribution as a prescribed series, sampled between their lower and upper bounds. Within each bound the rate declines linearly over 1901–1990, with a mean equal to their 1901–1990 value, and then holds their 1993–2010 rate. It does not depend on GMST and is set to zero from 2000, where GlaMBIE calibrates the glacier model. The first comparison is between global mean sea level and rate produced by SPECTRUM with data from *Fox-Kemper et al.* (2021) and *Frederikse et al.* (2020). For the second comparison, we calculate the transient sensitivity of the SPECTRUM output and GMSL data, where the transient sensitivity is defined as $\alpha = d\dot{S}/dT$ with $\dot{S}$ as the rate of change in GMSL. Here, we use the same rate-space blending scheme as the forecast (§2.2), reflected about the forecast origin so that the satellite-era quadratic dominates at late times and the component models at early times, with $t_r = 2015$ and $\tau_r = 5$ yr. This gives the observational term the same weight at 2025 that it carries at the forecast origin, and reduces it to 0.05 by 2000, so the hindcast over the reconstruction era tests the component models rather than restating the altimetry extrapolation.

Extending the hindcast back to 1900 shows distinct regimes relative to the various GMSL time-series (Fig. 8a). Over the satellite era the hindcast tracks the observations closely, with a 1993–2025 rate of 3.21 mm yr^{-1} against 3.35, 3.17, and 3.23 mm yr^{-1} for the *Frederikse et al.* (2020) and *Dangendorf et al.* (2024) reconstructions and the IPCC AR6 observational compilation (*Fox-Kemper et al.*, 2021), and with the satellite altimetry record (*Beckley et al.*, 2017). Over 1950–2000 the hindcast rate is 1.01 mm yr^{-1} against 1.02, 1.01, and 1.26 mm yr^{-1} for the same three records, and the hindcast level at 1950 sits 1–13 mm above them. Of the hindcast rate, 0.21 mm yr^{-1} comes from the prescribed uncharted-glacier term. Before 1950, where only the *Frederikse et al.* (2020) and *Dangendorf et al.* (2024) reconstructions extend, the divergence grows: the hindcast rate is 0.77 mm yr^{-1} against 1.39 and 1.64 mm yr^{-1} over 1900–1950, and the hindcast level at 1900 is 41–47 mm above the reconstructions, relative to the 2000 baseline. The median line from SPECTRUM falls outside the GMSL uncertainties of *Dangendorf et al.* (2024) only in 1900 and 1906–1909. The limitations of

SPECTRUM arise by design: we chose to focus only on model representations that can be fit to observations and that represent the ongoing response to warming. This choice excludes more complete and complex land-ice process representations that would be necessary to represent GMSL during colder periods in the past, as this is not our goal. We therefore treat the pre-1950 hindcast as qualitatively informative but do not draw quantitative conclusions from it. We restrict our validation claims to the 75-year, post-1950 period.

The transient sensitivity of our component model, which defines the change in the rate of change in GMSL with warming, tracks closely with the observed sensitivities (Fig. 8b). It is worth emphasizing that this agreement includes the transition from the earlier period when thermosteric sea-level rise was the largest contributor to GMSL to the present when the sum of land-ice and terrestrial water storage (barystatic) are larger overall contributors to GMSL. The transient sensitivity of the observations and model also agree with the semi-empirical results of *Rahmstorf* (2007) (Fig. 8b), which continue to track observations nearly 20 years after publication, showing that the simple semi-empirical model continues to provide a reasonable first-estimate projection of GMSL rise with warming.

Together, the level, rate, and transient-sensitivity comparisons indicate that our framework reproduces the observed behavior of GMSLR over a 75-year window (1950–2025) spanning 1.1–1.2 °C of GMST change, without having been calibrated on observations of GMSL itself. The hindcast reproduces the satellite-era rate (3.21 mm yr^{-1} over 1993–2025, against 3.17–3.35 mm yr^{-1} in the observational records). Its 1950–2000 rate of 1.01 mm yr^{-1} matches the two tide-gauge reconstructions (1.01 and 1.02 mm yr^{-1}) and falls below the 1.26 mm yr^{-1} of the IPCC AR6 compilation. Its transient sensitivity reproduces both the observed rate–temperature curvature and its underlying physical cause (the shift from thermosteric to barystatic dominance).

To our knowledge, no other GMSL forecasting framework or model presents this level of detailed validation or hindcast skill. Because this validation window is comparable in duration and magnitude of warming to the near- and mid-term forecast horizon, it supports extending the same confidence forward over a comparable span, through 2100 and a further 1–2 °C of warming. This validated envelope does not, however, extend to processes for which comparable historical constraints do not exist, which is why WAIS, the largest source of forecast uncertainty, is treated separately through a scenario-based 'storyline', rather than data-driven approach, in this work.

3.2 Component projections

We drive the calibrated model components with the IPCC AR6 GMST trajectories of 2 °C (SSP1-2.6), 3 °C (SSP2-4.5), and 4 °C (SSP3-7.0), the pathways most consistent with current policies and stated pledges (Fig. 9a). Current policies imply a median warming of 2.7 °C by 2100, with emissions that most closely match SSP2-4.5, although warming of 4 °C cannot be fully ruled out (*Hausfather*, 2025). Projected levels are for 2100 relative to the study baseline year 2000, given as posterior median with the 90% credible interval in brackets. Comparisons with the IPCC AR6 medium-confidence projections re-reference our projections to our model's 1995–2014 mean, the AR6 baseline (*Fox-Kemper et al.*, 2021).

The thermosteric contribution is 145 [107, 200], 244 [183, 340], and 370 [266, 537] mm at 2, 3, and 4 °C, respectively (Fig. 9b). The corresponding AR6 medians for thermal

expansion are 142, 197, and 249 mm. On the shared 1995–2014 baseline our 2 °C projection matches AR6, a ratio of 0.98. Our median then rises to 21% above the AR6 median at 3 °C and 46% above at 4 °C. Two features of the two-layer model (*Geoffroy et al.*, 2013) drive that divergence. The quadratic term in the upper-layer expansion carries the temperature dependence of the thermal expansion coefficient of seawater, so each added degree of warming expands the upper ocean more than the last. The deep layer also draws heat from the upper layer throughout the century, consistent with observed deep-ocean warming (*Purkey and Johnson*, 2010; *Cheng et al.*, 2017).

The Greenland Ice Sheet contributes 71 [35, 107], 93 [52, 132], and 117 [72, 163] mm at 2, 3, and 4 °C (Fig. 9c). The AR6 medians are 55, 83, and 114 mm. On the shared 1995–2014 baseline our medians are 1.24, 1.08, and 1.00 times the AR6 values. The independently calibrated Greenland posterior of *Aschwanden and Brinkerhoff* (2022) gives an envelope of 40 to 300 mm, which contains our 90% intervals at 3 and 4 °C. At 2 °C our 5th percentile of 35 mm lies just below it. Our Greenland discharge model is calibrated against the observed discharge record (*Mouginot et al.*, 2019), with two other mass-balance records withheld for validation (*Mankoff et al.*, 2021; *Otosaka et al.*, 2026). Our SMB emulator is trained on MAR alone, the regional climate model that gives the most SMB loss under the same forcing (*Glaude et al.*, 2024).

We project glacier contributions of 116 [99, 137], 139 [119, 170], and 161 [134, 197] mm at 2, 3, and 4 °C (Fig. 9d). The AR6 medians are 89, 123, and 157 mm. On the shared 1995–2014 baseline our ratios to AR6 are 1.27, 1.11, and 1.01, so our glacier projection converges on the AR6 median as warming increases. *Rounce et al.* (2023) report the glacier contribution over 2015–2100 relative to 2015, binned by 2100 warming. Our 2015 level is 9.8 mm above the 2000 baseline. Over that window we project medians of 106, 130, and 151 mm, against their ensemble medians of 99 ± 31, 125 ± 39, and 154 ± 44 mm, reported as median and 95% confidence interval. The agreement holds in all three scenarios. The rate of glacier mass loss is approximately linear with warming in the GlaMBIE record that we calibrate to (*The GlaMBIE Team et al.*, 2025), consistent with glacier model ensembles (*Rounce et al.*, 2023; *Marzeion et al.*, 2020). Our glacier warming-sensitivity prior is centered on 0.61 mm yr^{-1} $^{\circ}\mathrm{C}^{-1}$, drawn from *Rounce et al.* (2023), so this comparison is a consistency check. The prior on the sensitivity is weakly informative, and its posterior is set by the GlaMBIE record together with the 1850 balance constraint.

The combined Antarctic Peninsula and EAIS contribution stays near zero, with medians of −4.7 [−15.2, 3.8], −9.0 [−23.5, 2.9], and −13.0 [−31.0, 2.1] mm at 2, 3, and 4 °C (Fig. 9e). Negative values denote a sea-level fall. All three 90% intervals span zero, so the sign of the net contribution is not resolved. The EAIS contribution falls from −13.8 to −24.8 mm across the three scenarios as the accumulation response strengthens, partly offset by a Peninsula contribution rising from +8.9 to +11.6 mm. The EAIS projection adopts a literature surface-mass-balance sensitivity, $C_T = 60 \pm 20$ Gt yr^{-1} $^{\circ}\mathrm{C}^{-1}$ (*Frieler et al.*, 2015; *Ligtenberg et al.*, 2013). The IMBIE-3 regression serves as a diagnostic check on that value (*Otosaka et al.*, 2026). The Peninsula sensitivity is calibrated to the IMBIE-3 Peninsula record (*Otosaka et al.*, 2026). It reflects both surface melt and the dynamic response to ice-shelf collapse (*Scambos et al.*, 2004). West Antarctica is treated separately below.

3.3 Full model projections

For the 2 °C target, our forecast shows a median of 0.59 m [0.41–1.58 m] of GMSL relative to year 2000, which is 32% above the IPCC AR6 medium-confidence estimate of 0.44 m after referencing our projection to the AR6 baseline. Under 3 °C warming, approximately the current trajectory (*Hausfather*, 2025), the median is 0.74 m [0.53–1.72 m], 30% above IPCC's 0.56 m. For the 4 °C trajectory, the median reaches 0.93 m [0.68–1.88 m], 35% above IPCC's 0.68 m. For reference, the 5 °C warming trajectory (SSP5-8.5) yields a median of 1.10 m [0.77–2.06 m].

Relative to pre-industrial GMSL (adding ∼0.19 m of observed rise through 2000), these forecasts correspond to medians of 0.78–1.12 m of global mean sea level rise across the 2-4 °C policy-relevant pathways. The probability of exceeding 1 m relative to pre-industrial by 2100 is 24% under 2 °C warming, 39% under 3 °C, and 73% under 4 °C warming trajectory. The probability of exceeding 2 m relative to pre-industrial by 2100 ranges from 4% to 6% across the 2–4 °C pathways, reaching its minimum if warming is held below 2 °C. Where slow WAIS mass loss is very likely, the probability of exceeding 1 m relative to pre-industrial by 2100 falls to 3% at 2 °C and 5% at 3 °C, though it remains 37% at 4 °C, and the probability of exceeding 2 m stays below 1% across the 2–4 °C pathways (Fig. 10b). This implies lower flood risks for tens of millions of people and reduced costs of adaptation and damages of order trillions of US dollars (Fig. 11a). As validation, our component forecast for 3 °C of warming is within 7 cm of the extrapolations of observed GMSLR trends, as should be expected if the warming trend remains similar to the current trend and WAIS, which has contributed 9% of observed GMSLR over the satellite altimetry record (1993–2024), continues to lose mass at its current rate.

Across the range of possible WAIS mass loss and emission scenarios, our forecasts suggest that WAIS is likely to become the dominant source of sea-level rise before the end of the century (Fig. 11a). It is uncertain when that will happen and depends on how quickly emissions decrease. The stack of mean values shows the relative contributions of the primary sea-level contributors, showing that the thermosteric component likely will remain the single largest contributor in the 21st century unless WAIS mass losses overtake it. Greenland and mountain glaciers will continue to lose mass and contribute to sea level rise throughout the 21st century.

The distribution of the uncertainties shows a dramatic separation between the predictable and, currently, unpredictable components (Fig. 11b). These results show that WAIS accounts for more than 90% of the variance, and 73 to 85% of the 90% credible interval, in 21st century sea-level rise. The thermosteric, glacier, and Greenland contributions, all of which have calibrated sensitivities to global mean surface temperature, contribute 1 to 7% to the total variance and 15 to 27% to the 90% credible interval, with their share of that interval growing as warming increases. As a result, the 90% uncertainty range of 1.18 m for 3 °C is 3.5 times larger than the spread across warming scenario medians (0.34 m from 2–4 °C). This picture shifts where slow mass loss is very likely. Those three components then set 43 to 61% of the 90% credible interval, up from 15 to 27% (right column of Fig. 11b). The relative magnitude of uncertainties reveals that sea-level risk decomposes into two approximately orthogonal dimensions, each addressed by a distinct class of societal response.

4 Discussion

4.1 Data-driven forecasts as part of a hierarchy

The SPECTRUM framework occupies an intermediate level of a hierarchy of sea-level models ordered by complexity, allowing us to compare the results with the simpler and the more complex models. The simplest models extrapolate the observed trends. Carried forward to 2100, a quadratic fit to the satellite altimetry measurements of GMSL gives 0.66 [0.41, 0.91] m of rise relative to the year 2000 (*Hamlington et al.*, 2024; *Nerem et al.*, 2022). Extrapolation also holds up at the local scale over the coming decades. Trends and accelerations at 222 tide gauges over 1970–2023, extrapolated to 2050, agree with the AR6 medium-confidence projections within the likely range at 96% of locations (*Wang et al.*, 2025). The next level of model complexity are the semi-empirical models that fit the rate of global mean sea-level rise to GMST. Applying the semi-empirical relation of *Rahmstorf* (2007) to the current warming trajectory gives 0.67 [0.63, 0.71] m. The calibrated SPECTRUM projection in which WAIS mass loss is very likely to contribute along the observed trajectory gives 0.59 [0.46, 0.82] m at 3 °C (Fig. 10a). The median is 7 cm below the naive extrapolation of GMSL and 8 cm below the semi-empirical model. Overall, the agreement to within 8 cm between the calibrated SPECTRUM model — forced with a warming scenario that approximates the current warming trend — and the simpler, empirical approaches provides an additional layer of comparison for the forecast.

At the complex end of the hierarchy are the process-model assessments of the IPCC AR6, which report two sets of projections. The medium-confidence set is built from processes that can be modeled with confidence, while the low-confidence set admits ice sheet processes such as marine ice cliff instability along with expert judgment. In SPECTRUM, the value p(S1) scales between these confidence intervals. Referenced to the AR6 1995–2014 baseline, which sits 1.2 cm above the year-2000 baseline used above, our slow-WAIS forecast gives 0.44 [0.33, 0.63] m at 2 °C, 0.58 [0.45, 0.81] m at 3 °C, and 0.76 [0.58, 1.05] m at 4 °C, against AR6 medium-confidence medians of 0.44, 0.56, and 0.68 m. Our medians run 1 to 11% above those values rather than the 30 to 35% of the full forecast, and our upper bound stays inside the AR6 range at all three warming levels (0.63, 0.81, and 1.05 m against 0.78, 0.95, and 1.11 m). Our fast-WAIS forecast tracks the low-confidence projections, which AR6 provides for SSP1-2.6, SSP2-4.5, and SSP5-8.5. At 3 °C our median of 0.72 m and 95th percentile of 1.71 m sit above the low-confidence values of 0.57 and 1.18 m, while at 5 °C our 95th percentile of 2.05 m falls below theirs of 2.28 m. Most of the difference between our forecast and AR6 is therefore a difference in the weight given to rapid WAIS mass loss, not in the temperature-sensitive components.

4.2 Completeness of the uncertainty budget

Aschwanden et al. (2021) separate the uncertainty in an ice-sheet forecast into five parts: the emissions scenario, the model structure, the initial state, the model parameters, and the irreducibly random component of the forcing. Their criticism of the ISMIP6 ensemble is that it integrated over model structure but not over parameters, in a setting where the parametric spread is the larger of the two. SPECTRUM does the converse. It integrates

over parameters for every calibrated component, and over structure for the one component where structure dominates, through the WAIS scenario mixture. Within-scenario forcing uncertainty, the spread of GMST trajectories under a given emissions pathway, is sampled explicitly as a warming path shared by all components in each ensemble member. Initial-state and random-forcing uncertainty enter through the calibrated error models rather than being separated. What the intervals do not carry is structural uncertainty in the calibrated components themselves, in the sense of alternatives to the functional form each component takes. For the full forecast this has little consequence, because WAIS accounts for more than 90% of the variance. Where slow WAIS mass loss is very likely, the calibrated components account for 43 to 61% of the summed component confidence interval widths, and their contribution to the total confidence interval is conditional on the model forms adopted here. Two of those choices have been tested directly. Substituting the observed Greenland-to-global warming ratio for the adopted Arctic amplification factor, and averaging the ocean forcing over a shallower or a deeper layer, each move the 2100 median by 4 mm or less and leave the 90% credible interval unchanged. *Lin et al.* (2026) integrate over the ice-model choices that ISMIP6 documented but did not sample. They emulate the ISMIP6-2300 ensemble as a function of 18 ice-dynamical features together with the choice of climate model, then place a uniform prior over those choices. In their raw simulations these features account for 64% of the 2015–2100 variance, with the sliding law contributing 21% and the ice-shelf melt parameterization 20%, compared with 9% for the choice of atmosphere-ocean general circulation model and 6% for GMST. The spread arising from choices made within the models is therefore several times the spread arising from the forcing.

We do not calculate the integral over model structure itself, in which a projection is averaged over a distribution across model structures (*Aschwanden et al.*, 2021). That distribution is defined by the requirement that the true data-generating process lie within its support. Therefore the distribution cannot be recovered by assigning weights to the models that have been built and requiring those weights to sum to one. Doing so treats each model as an independent sample of the ways the system might be represented, when in fact those models describe the same system, rely on many of the same assumptions, and are tested against many of the same observations. What the SPECTRUM framework does implement is the conditioning step of *Aschwanden et al.* (2021), in which a prior over projections is filtered by observations. Observed GMSL is withheld from every component calibration, so the hindcast tests the assembled framework against a record none of its components saw. Over 1950–2000 the summed components rise at 1.01 mm yr^{-1} against 1.02 and 1.01 mm yr^{-1} in the *Frederikse et al.* (2020) and *Dangendorf et al.* (2024) GMSL data sets. *Aschwanden et al.* (2021) point out that it is not yet settled how long a model should be required to match observations before it is trusted. The test used here covers the 75 years since 1950, which is longer than the mass-change records such comparisons usually rely on, although the reconstructions it is compared against also become less certain further back in time. *Lin et al.* (2026) take the alternative approach for Antarctica, assigning probabilities to the ISMIP6 modeling options and updating them against the observed mass-loss rate. Their result illustrates the limit of that approach as well as its value. The observations move the sliding-law probabilities, but for most other choices the posterior stays close to the prior, because a two-decade mass-balance record weakly constrains processes that have little effect on the recent trend and a large effect on the century-scale response. They describe the

residual as strong ambiguity and address it with probability boxes over alternative priors, the same difficulty that leads us to represent WAIS with a scenario mixture rather than a single distribution.

4.3 Orthogonal dimensions of sea-level risk

The potential impact and uncertainty around WAIS mass loss creates two orthogonal dimensions of risk. The first risk dimension is sensitive to emissions reduction. The temperature-dependent components are sensitive to changes in GMST. Here, we calibrate the sensitivities using decades of observations and validate the fit against decades of observations withheld from the calibration. The collective contribution of the calibrated components shifts systematically across warming trajectories and is reducible by emissions cuts (Fig. 11a). Under the 'slow WAIS' scenario, the 90% credible interval of the forecast spans 31 cm at 2 °C, 36 cm at 3 °C, and 46 cm at 4 °C, 2.6 to 3.8 times narrower than the full forecast that admits rapid mass loss from WAIS driven by internal feedbacks (Fig. 10). Emissions policy becomes the primary lever for managing risk in that world. Moving from 2 °C to 4 °C adds 31 cm of rise, exposing 52 million more people to coastal flood risks and adding US$42B/yr in coastal adaptation costs.

The second risk dimension is relatively insensitive to emissions reductions over the 21st century. WAIS dynamics accounts for more than 90% of the total forecast variance and is largely decoupled from the emission pathway. The tail of the forecast distribution—the outcomes that drive the most severe human and economic consequences—is controlled almost entirely by whether WAIS retreat follows modern loss trends or transitions to a more rapid, irreversible retreat (*Turney et al.*, 2020; *Dutton et al.*, 2015; *Graham et al.*, 2022). In this case, emissions reduction shifts the whole distribution downward, lowering the median and the 95th percentile by similar amounts, but does not narrow the upper tail, whose spread above the median (0.99 and 0.95 m at 2 and 4 °C) is set by WAIS. The probability of exceeding 2 m relative to pre-industrial is 4% under the 2 °C trajectory and 6% under 4 °C. The 4% lower bound on this range highlights the fact that emission reductions alone cannot retire the high risk outcomes, though it is predicated on the values in the AR6 low-confidence scenarios that arise more from structured expert judgment than from data and reliable physical models. Emerging research on processes absent from current ice-sheet models points toward mechanisms that could raise those values (Section 2.2.2).

The risks and costs to coastal communities helps to highlight the dichotomy between the two possible scenarios of fast vs slow WAIS mass loss, or equivalently, emissions-insensitive vs emissions sensitive hazards (Figs. 10 and 11a). The difference at the 5% exceedance probability, one possible design threshold for coastal adaptation (*van de Wal et al.*, 2022) (Fig. 10b), is 0.89 m of sea-level rise by 2100 under the current policy warming trajectory (3 °C). This difference translates into roughly 170M more people exposed to annual flood risks, US$120B/yr in coastal adaptation, and US$12.2T/yr (about 10% of global GDP per year) in coastal flood damages without further adaptation. For intuition, the fast-WAIS scenario at 5% exceedance implies more than seven times more sea-level rise in the next 75 years than was experienced in the past 125 years, while the slow-WAIS scenario implies more than three times the sea-level rise.

Resolving the question of WAIS mass loss rates is among the most financially consequen-

tial goals in modern science (Fig. 11c). A value-of-information analysis shows that knowing whether WAIS will lose mass slowly or rapidly is worth roughly US$4.5 trillion in avoided costs. That figure is the maximum over the range of prior confidence in slow WAIS mass loss, reached at 82% confidence for information arriving today. At the p(S1) = 90% case adopted here the value is US$3.9T. The value decays with time because late-arriving information leaves fewer years for decisions to improve. These numbers frame the research investment question in concrete terms: even partial reduction of WAIS uncertainty, well short of perfect knowledge, would justify annual research expenditures orders of magnitude above current levels, $\sim$ $10M *Catania et al.* (2022). The return on that investment is determined by how quickly the observations can narrow the range of WAIS outcomes, which depends on both the rate of data collection and the development of actionable process-based model results, like those used in the IPCC, capable of providing explanatory insights that underscore deeper understanding of ice sheets and climate.

Scientific research is a necessary condition for providing actionable information for coastal adaptation, but it is as likely as not that further research will add to the growing evidence to support rapid WAIS mass loss rather than build confidence in a slow-WAIS future (*Fricker et al.*, 2025). Sustaining a wide range of outcomes that admit $\leq$ 0.72 m and $\geq$ 1.91 m of GMSLR in 2100 at 5% probabilities, will force decisions on coastal planning, adaptation, and managed retreat that impact millions of lives and cost billions of dollars to be made under large uncertainties. The extent of these uncertainties in terms of lives, property, and ecosystems raises the question: Can we take action to increase the potential stability of West Antarctica? How much harm to people and the environment can we prevent? Recently, some research directions have been proposed to answer this question (*MacAyeal et al.*, 2024) with initial models of glacier interventions aimed at stabilizing WAIS show promise (*Meyer et al.*, 2026; *Miller et al.*, 2026). In the context of this study, stabilizing WAIS means increasing $p(S1)$ and reducing the 95% (5% exceedance) of S2, the scenario that admits rapid WAIS mass loss. As with any nascent research, there are many unresolved questions about the feasibility, unintended consequences, and governance of these interventions (*Siegert et al.*, 2025). There are risks in intervening to remediate the West Antarctic Ice Sheet and slow its loss. Similarly, the logistical challenges will undoubtedly be great, but these risks and challenges should be weighed against the risks and challenges of coastal adaptation for sea-level rise. There is no response to sea level rise that comes with zero risk and zero cost.

5 Conclusions

We focus on projecting 21st-century sea-level rise with an intermediate-complexity model in which calibration, validation, and uncertainty quantification are primary features. SPECTRUM represents each contributor with a parsimonious physical model, including a new ocean-forced model of Greenland outlet-glacier discharge. Each is calibrated by Bayesian inference against independent observational records where such records exist, and emulated from regional climate models or adopted from published sensitivities where they do not. With observed GMSL withheld from every calibration, the summed components close 93–101% of the satellite-altimetry rate. They reproduce the level of multiple independent reconstructions within error since approximately 1950 and 1.2 °C of warming, and they reproduce the

observed sensitivity of the rate of rise to warming, including the shift from thermosteric to barystatic dominance, and the 1950–2000 rate of the tide-gauge reconstructions. Because the validation window is comparable in duration and warming to the forecast horizon, this hindcast skill supports forecasting the calibrated components through 2100. Where WAIS is assumed very likely to continue its observed mass-loss trajectory (p(S1) = 90%), SPECTRUM at 3 °C gives a median of 0.59 m, 7 to 8 cm below the extrapolation of observed GMSL trends (0.66 m) and semi-empirical models (0.67 m). It projects medians of 0.45 and 0.77 m at 2 °C (SSP1-2.6) and 4 °C (SSP3-7.0), 1–11% above AR6 medium confidence. Under an assumed weighting of 90% toward accelerated WAIS loss (p(S1) = 10%), the medians rise to 0.59, 0.74, and 0.93 m. The 90% credible interval at 3 °C widens from [0.46, 0.82] m to [0.53, 1.72] m.

Under the accelerated weighting, the uncertainty in these forecasts has two sources. The first is the set of contributors whose temperature sensitivity we calibrated against observations. These account for 1 to 7% of the forecast variance and 15 to 27% of the width of the 90% credible interval. Their contribution shifts systematically with warming, and emissions cuts reduce it. The second is the West Antarctic Ice Sheet, which accounts for more than 90% of the forecast variance and for 73 to 85% of the credible interval. WAIS responds weakly to the emissions pathway. Holding warming to 2 °C rather than 4 °C lowers the median by 0.34 m but moves the probability of exceeding 2 m above pre-industrial only marginally, from 6% to 4%, a change that is small relative to the weak physical constraints on the tail. Beyond 2100 the picture changes, because 21st-century warming sets the long-term commitment from Antarctica (*Lowry et al.*, 2021; *Seroussi et al.*, 2024). Under the slow weighting, the calibrated contributors set 43 to 61% of the interval.

The two sources of uncertainty call for different responses. Emissions cuts reduce the relatively predictable, temperature sensitive part of sea-level rise, which includes the thermosteric, glacier, and Greenland Ice Sheet contributions. Resolving the rate of West Antarctic mass loss, which recent studies have found to be relatively insensitive to emission reductions this century, would narrow the range that coastal communities must prepare for. We estimate that information to be worth up to about US$4.5 trillion in avoided damages and adaptation costs, a value that declines the later the information arrives. The rate of West Antarctic ice loss over the next two decades is likely the most informative measurement, along with more detailed and persistent monitoring of key areas of West Antarctica to inform forecast models. The cost of such monitoring is paltry compared with the cost of not resolving the question of West Antarctic mass loss. However, emissions cuts and better data and models may not reduce the 21st-century tail risk of rapid sea-level rise, although emissions cuts do reduce the long-term commitment from West Antarctica (*Lowry et al.*, 2021). Whether interventions aimed at slowing West Antarctic ice loss could do so is an open question.

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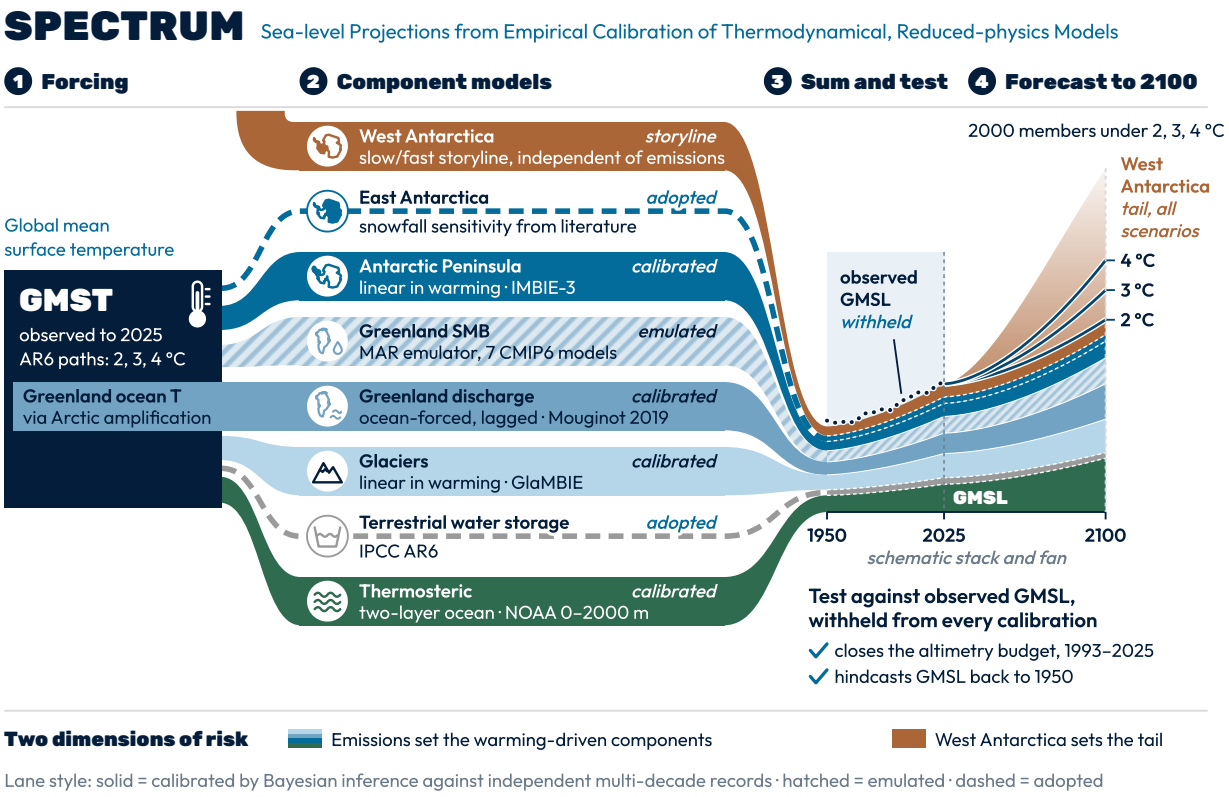


Figure 1: **The SPECTRUM framework.** (1) Global mean surface temperature (GMST), observed to 2025 and following AR6 paths that end near 2, 3, and 4 °C, forces each component. Greenland discharge is forced by ocean temperature around Greenland, derived from GMST through Arctic amplification. (2) Each contributor has its own model. Line style marks how the model is constrained: solid lanes are calibrated by Bayesian inference against independent multi-decade records, the hatched lane is emulated, and dashed lanes are adopted from published sensitivities and assessments. West Antarctica (brown) follows a slow/fast storyline that is independent of emissions this century. (3) The components are summed and tested against observed GMSL, which is withheld from every calibration. (4) The forecast to 2100 uses 2000 members under each warming level. Emissions set the warming-driven components, and West Antarctica sets the tail in all scenarios. The stacked areas and fan are schematic.

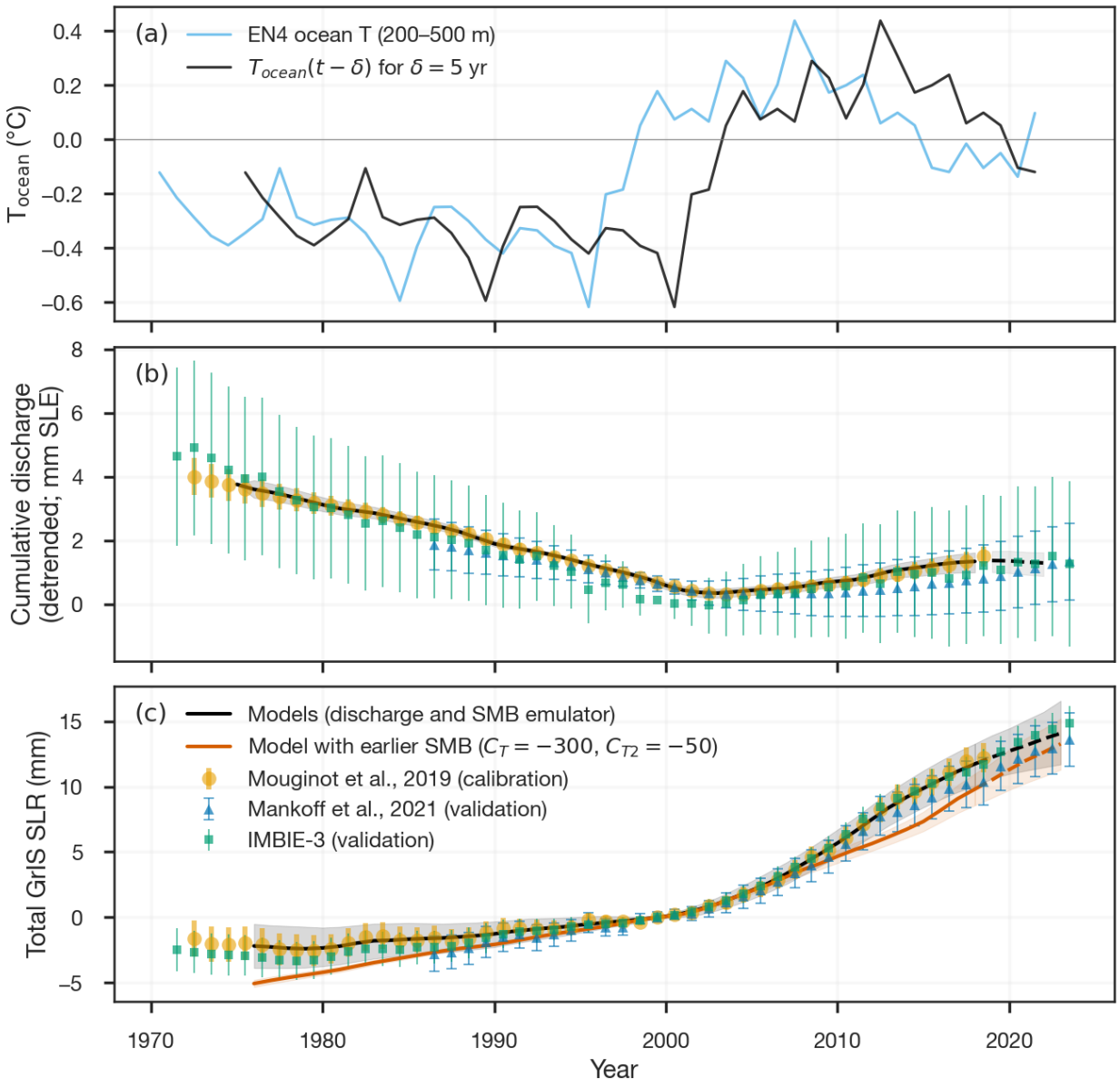


Figure 2: **Greenland discharge delay model fit.** **(a)** Subsurface ocean temperature (from EN4 averaged over depths of 200–500 m with annual resolution; blue) and (black) $T_{\text{ocean}}(t-\delta)$, the same signal delayed by $\delta = 5$ yr. **(b)** Detrended cumulative Greenland ice discharge showing model fit (black) with 90% posterior CI, compared with the calibration data (*Mouginot et al.*, 2019) (orange) and the time-series of *Mankoff et al.* (2021) (blue) and IMBIE-3 (*Otosaka et al.*, 2026) (green) used for validation. **(c)** Total Greenland mass balance, the sum of modeled SMB and modeled discharge (black, with 90% interval). The SMB is the MAR emulator with the drift state fitted to the *Mouginot et al.* (2019) SMB, so over 1972–2018 this panel tests the discharge model. Dashed lines continue the models past the end of the *Mouginot et al.* (2019) record. For comparison, the orange line replaces the SMB with a parametric model, $380 + C_T\,\Delta T + C_{T^2}\,\Delta T^2$ Gt yr^{-1} with $C_T \sim \mathcal{N}(-300, 80^2)$ Gt yr^{-1} °C^{-1} and $C_{T^2} \sim \mathcal{N}(-50, 30^2)$ Gt yr^{-1} °C^{-2}, where ΔT is the GMST anomaly relative to 1995–2005 and both terms are zero when $\Delta T < 0$.

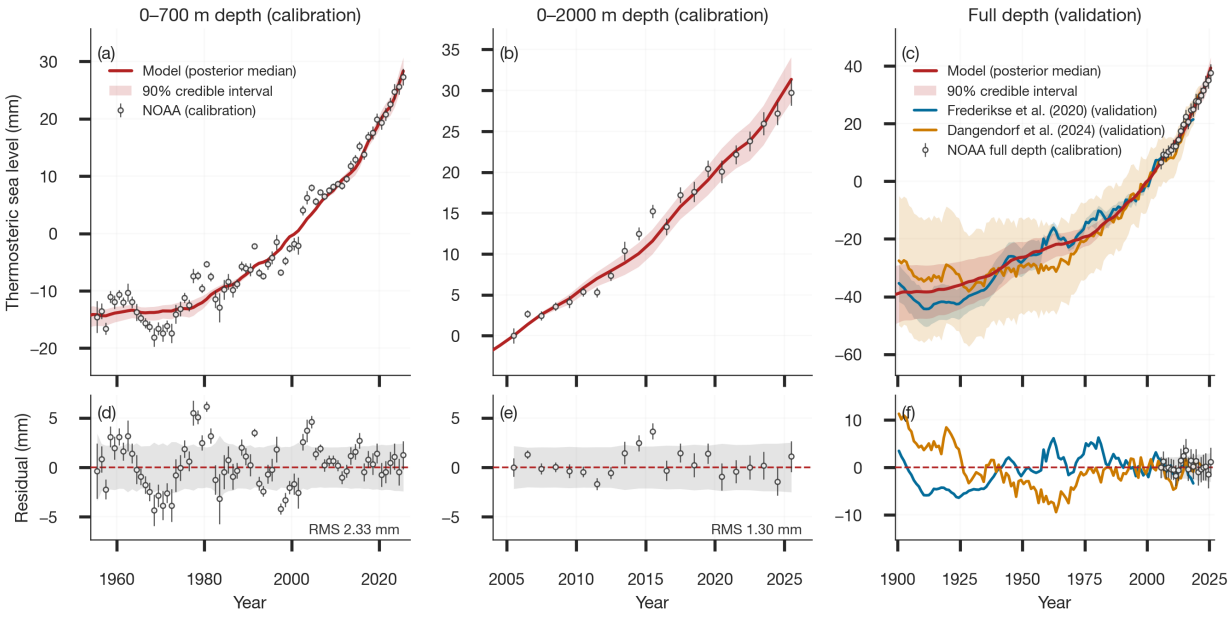


Figure 3: **Two-layer thermosteric model fit and validation.** (a) Posterior median of the upper layer model H_{u} (Eq. 8) and 90% credible interval fitted to observations at 0–700 m depth (circles with error bars; from NOAA (*Levitus et al.*, 2012)). (b) Same as in (a) but for the 0–2000 m data and H_{ud} (Eq. 11). (c) Validation of the full-depth model H_{full} (Eq. 12) against the results from *Frederikse et al.* (2020) (blue) and *Dangendorf et al.* (2024) (orange). Also shown for reference are the full-depth values (circles, same as in panel b). (d–f) Residuals corresponding to (a)–(c), with the gray band showing the 90% credible interval on the residual and the RMS residual reported in each calibration panel.

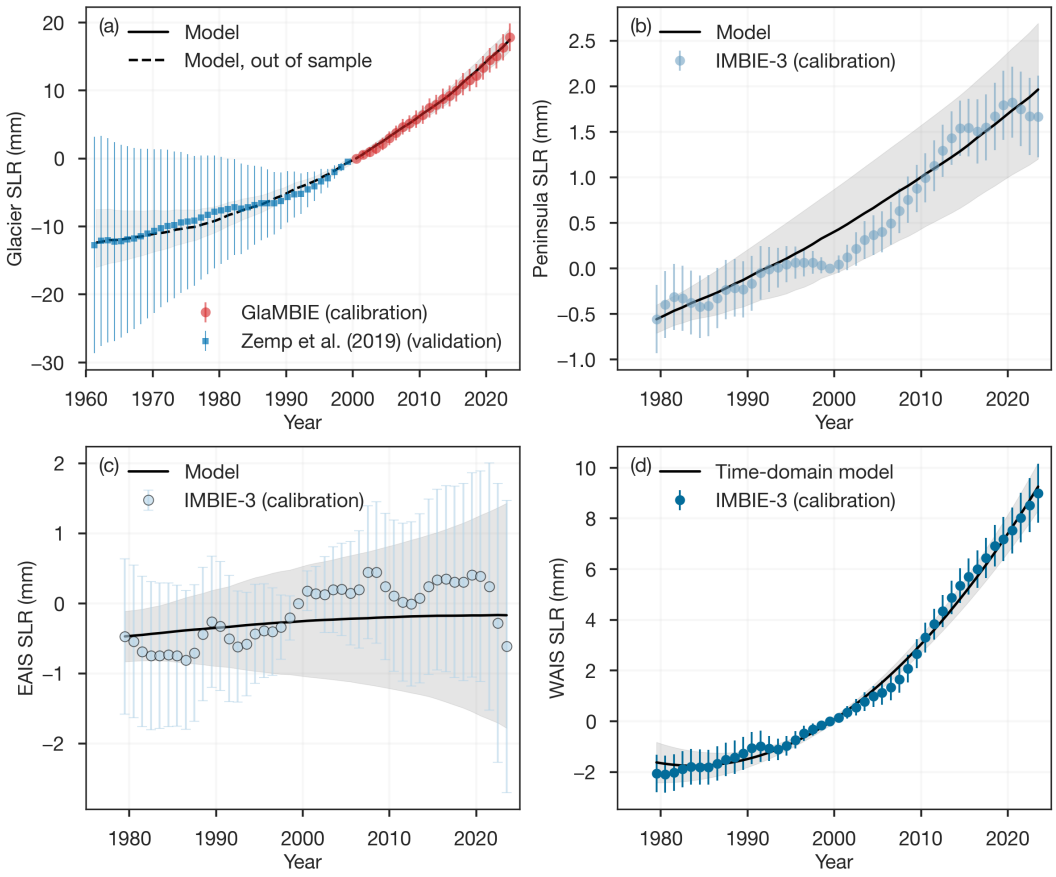


Figure 4: **Per-component calibration fits.** **(a)** Glaciers, **(b)** Antarctic Peninsula, and **(c)** EAIS observed mass balance (circles with ±2 standard deviation error bars) with the respective posterior median rate–temperature models (black lines) and 90% credible interval (gray shading). **(d)** WAIS mass balance is shown with the time-domain model (black line) and 90% credible interval (gray shading). In (a) the glacier model is also run back to 1961 (dashed line and lighter shading, outside the calibration period) and compared with the validation record of *Zemp et al.* (2019) (squares), whose cumulative values are matched to the model at the end of 1999. Data are from *The GlaMBIE Team et al.* (2025) (calibration) and *Zemp et al.* (2019) (validation) for glaciers and *Otosaka et al.* (2026) for the ice sheets.

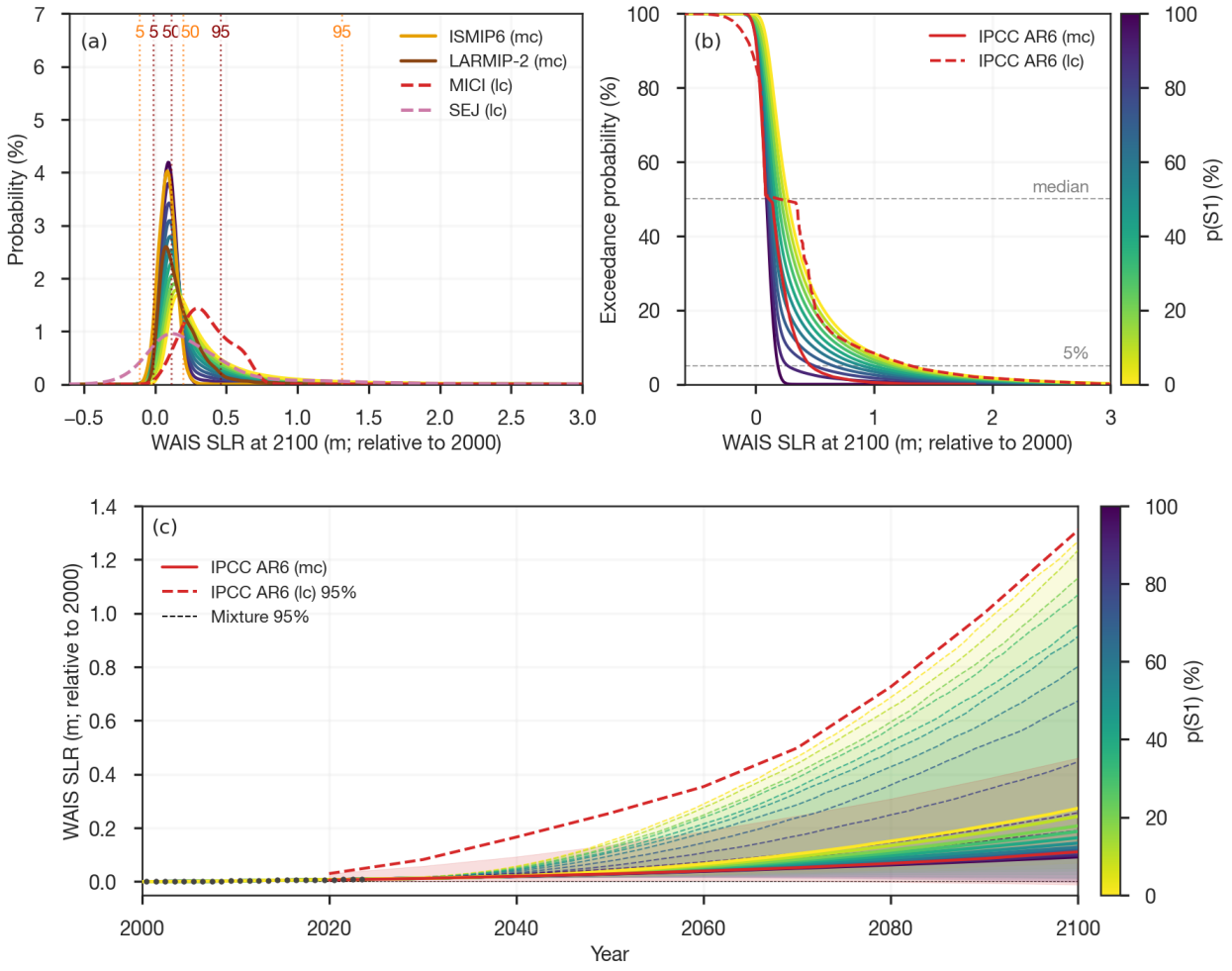


Figure 5: **WAIS 'storyline' scenario mixture model. (a)** Probability density of the WAIS contribution to GMSL at 2100 (relative to 2000) for an 11-member sweep of the mixture weight p(S1) from 100% (Scenario 1) to 0% (Scenario 2), colored by weight (colorbar). Also shown are the IPCC AR6 constituent model ensembles for the Antarctic Ice Sheet workflow underlying the medium-confidence (mc, ISMIP6 emulator (*Edwards et al.*, 2021) and LARMIP-2 ensemble (*Levermann et al.*, 2020)) and low-confidence (lc) assessments. The low-confidence assessments are identified as Marine Ice Cliff Instability (MICI) (*DeConto and Pollard*, 2016) and structured expert judgement (SEJ) (*Bamber et al.*, 2019). Dotted vertical lines mark the 5th, 50th, and 95th percentiles of the medium- (red) and low- (orange) confidence AR6 p-boxes. **(b)** Corresponding exceedance probability curves for the same p(S1) sweep, with the IPCC AR6 medium for 3 °C (SSP2-4.5; red solid line) and low-confidence SSP5-8.5 (red dashed line) exceedance curves for the Antarctic Ice Sheet overlaid for reference. **(c)** Time-series for the p(S1) sweep and the IPCC AR6 medium SSP2-4.5 (red solid line with 90% CI shading) and low-confidence SSP5-8.5 95th percentile (red dashed line). The thin dashed lines show the 95th percentile for the correspondingly colored $p(S1)$.

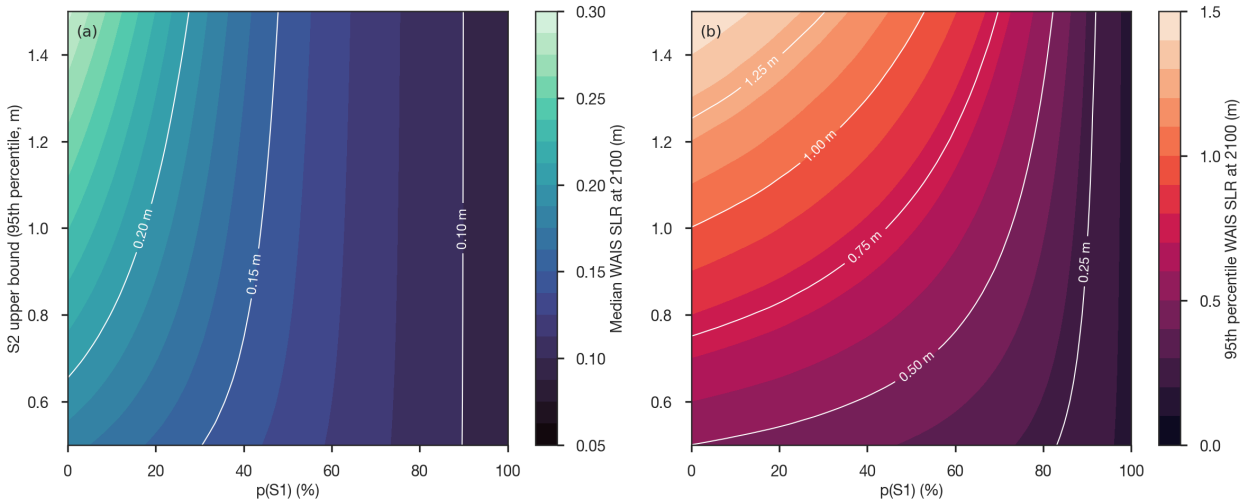


Figure 6: **WAIS contribution to GMSL at 2100 as a function of the two mixture parameters. (a)** Median and **(b)** 95th percentile of the S1/S2 mixture (relative to 2000) across the S1 probability, p(S1), and the S2 upper bound (95th percentile of S2 at 2100).

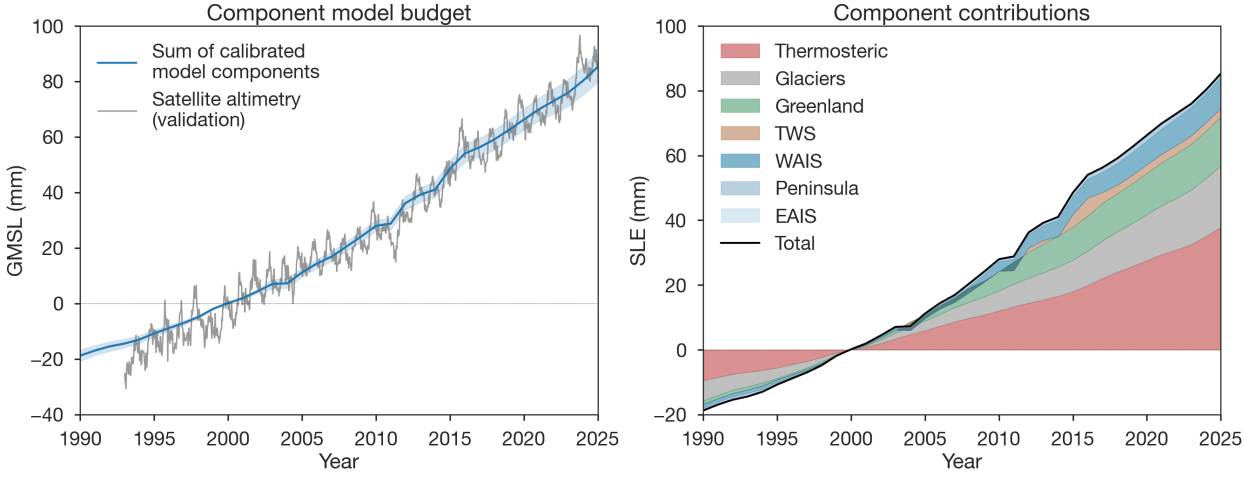


Figure 7: **Budget closure (validation): sum of model components versus observed GMSL. (a)** Sum of components (blue) and NASA satellite altimetry (grey) with shaded band showing the 90% confidence intervals (*Beckley et al.*, 2017; *Hamlington et al.*, 2020). **(b)** Stacked per-component contributions and their sum in black. Terrestrial water storage enters only from 2003, the start of the GRACE record from which it is derived.

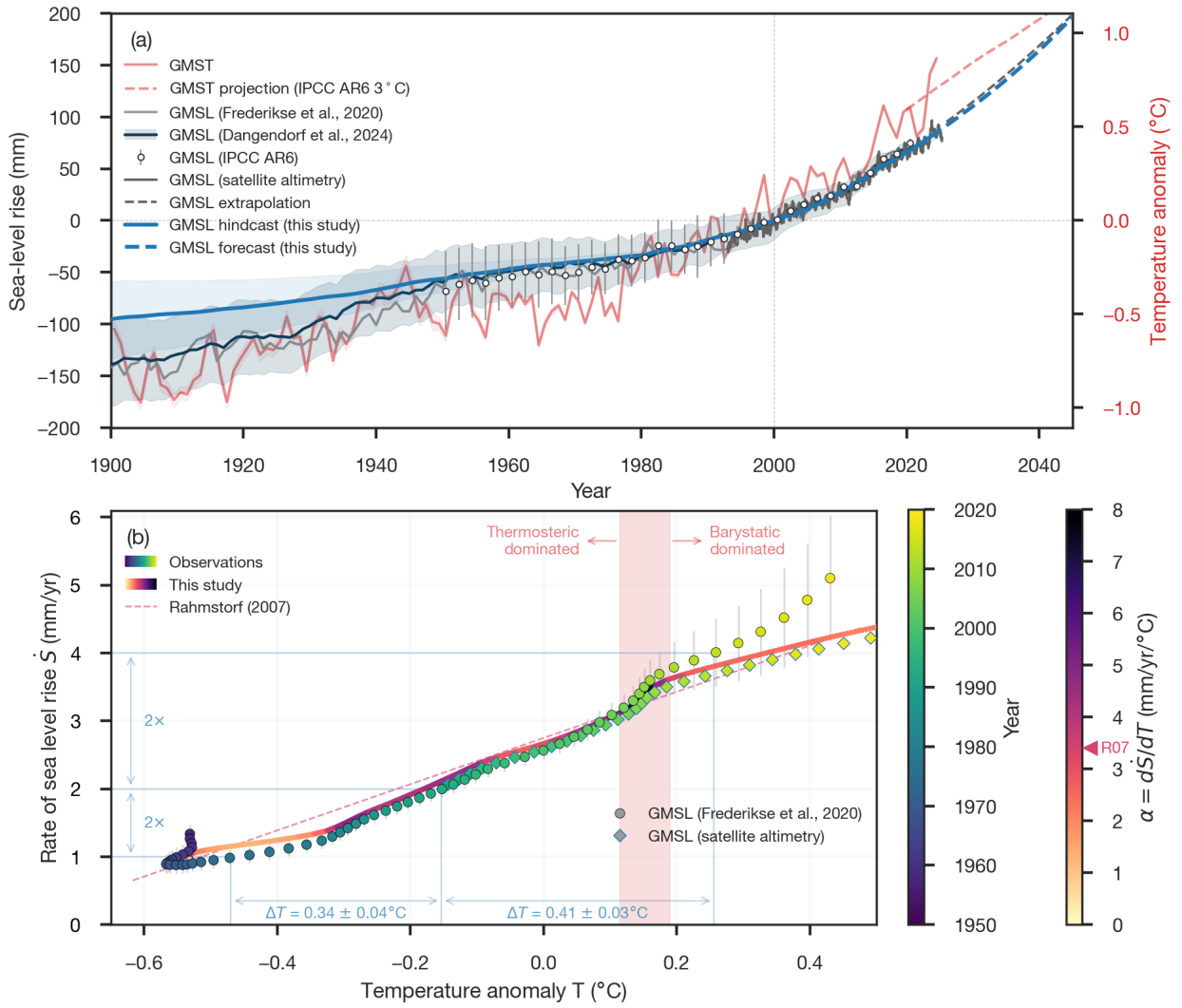


Figure 8: **Global mean sea-level rise and global mean surface temperature. (a)** GMSL reconstructions from *Frederikse et al.* (2020) (light grey) and *Dangendorf et al.* (2024) (navy, with 90% CI shading), the IPCC AR6 observational compilation (*Fox-Kemper et al.*, 2021) (white circles with 90% CI bars), and NASA satellite altimetry (*Beckley et al.*, 2017) (mid grey, with its quadratic extrapolation dashed) show sea level rising with warming. The hindcast and forecast of this study are shown in blue, solid and dashed respectively, with 90% CI shading. Before 2000 the hindcast includes the prescribed contribution of uncharted glaciers (*Parkes and Marzeion*, 2018). GMST (red; (*Rohde and Hausfather*, 2020)) is referenced to the right hand axis. **(b)** Rate of GMSL rise ($\dot{S} = dGMSL/dt$) versus GMST anomaly, T. Observed values are shown in circles (*Frederikse et al.*, 2020) and diamonds (*Beckley et al.*, 2017), colored by year, and the component model of this study is a line colored by instantaneous transient sensitivity $\alpha = d\dot{S}/dT$.

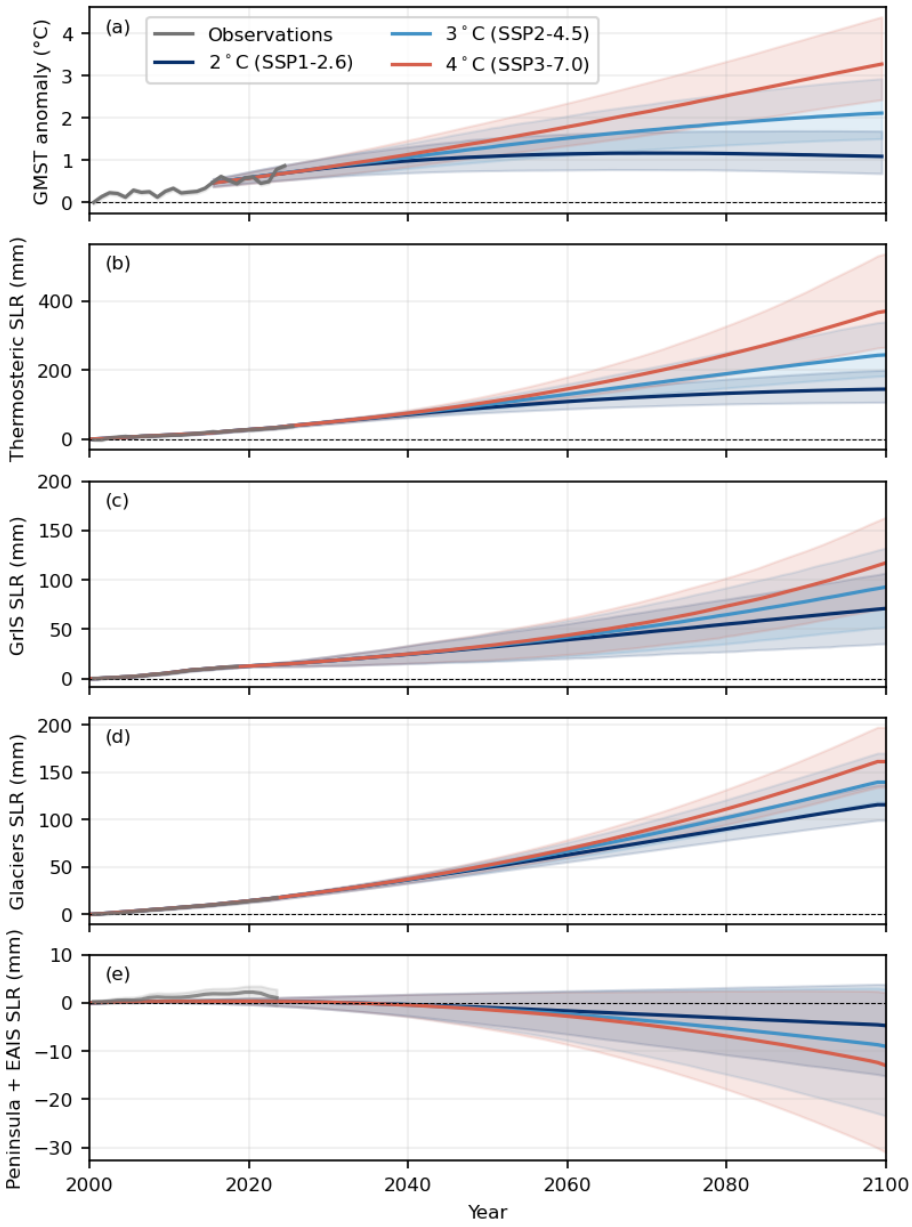


Figure 9: **Component-level forecasts under three emissions scenarios.** (a) Observed global mean surface temperature (GMST) anomaly (grey) relative to 2000, with SSP1-2.6, SSP2-4.5, and SSP3-7.0 projections (IPCC AR6) and their 90% CI (5–95th percentile, shaded). The SSP projections are labeled in the legend by their median 2100 anomaly relative to pre-industrial temperatures to provide both pre-industrial and 21st century baselines. (b) Thermosteric sea-level contribution with observations (grey) and forecast median with 90% CI from the full depth model driven by the corresponding GMST trajectory. (c) Greenland Ice Sheet (GrIS), (d) glaciers, and (e) the sum of Antarctic Peninsula and EAIS following the same convention as in (b).

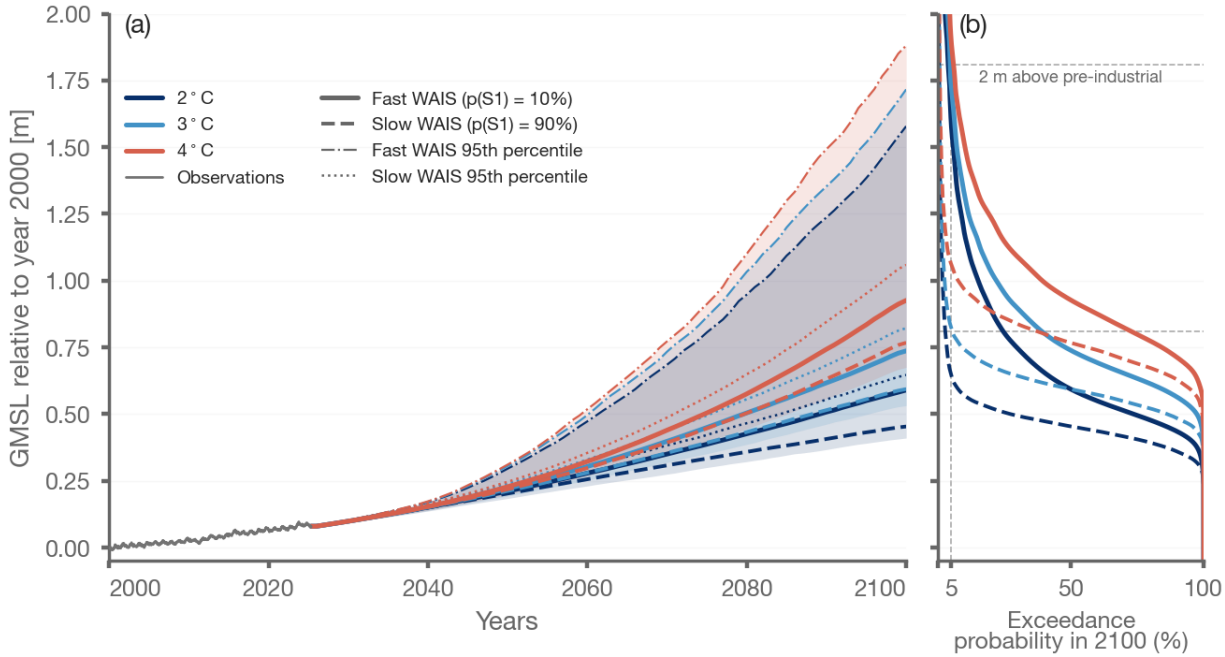


Figure 10: **Global mean sea-level projections.** (a) Global mean sea level relative to 2000 under 2 °C (SSP1-2.6), 3 °C (SSP2-4.5), and 4 °C (SSP3-7.0) of warming, with satellite altimetry observations (grey) (*Beckley et al.*, 2017). Solid lines show the fast WAIS forecast median with its 90% CI (5–95th percentile, shaded). Dashed lines show the slow WAIS median. Dash-dot and dotted lines mark the 95th percentile of the fast and slow WAIS forecasts, respectively. (b) Probability of exceeding each sea level in 2100, on the same sea-level axis as (a), for the fast (solid) and slow (dashed) WAIS forecasts. Horizontal dashed lines mark 1 m and 2 m above pre-industrial, and the vertical dashed line marks 5% exceedance, the level traced over time by the 95th percentile lines in (a).

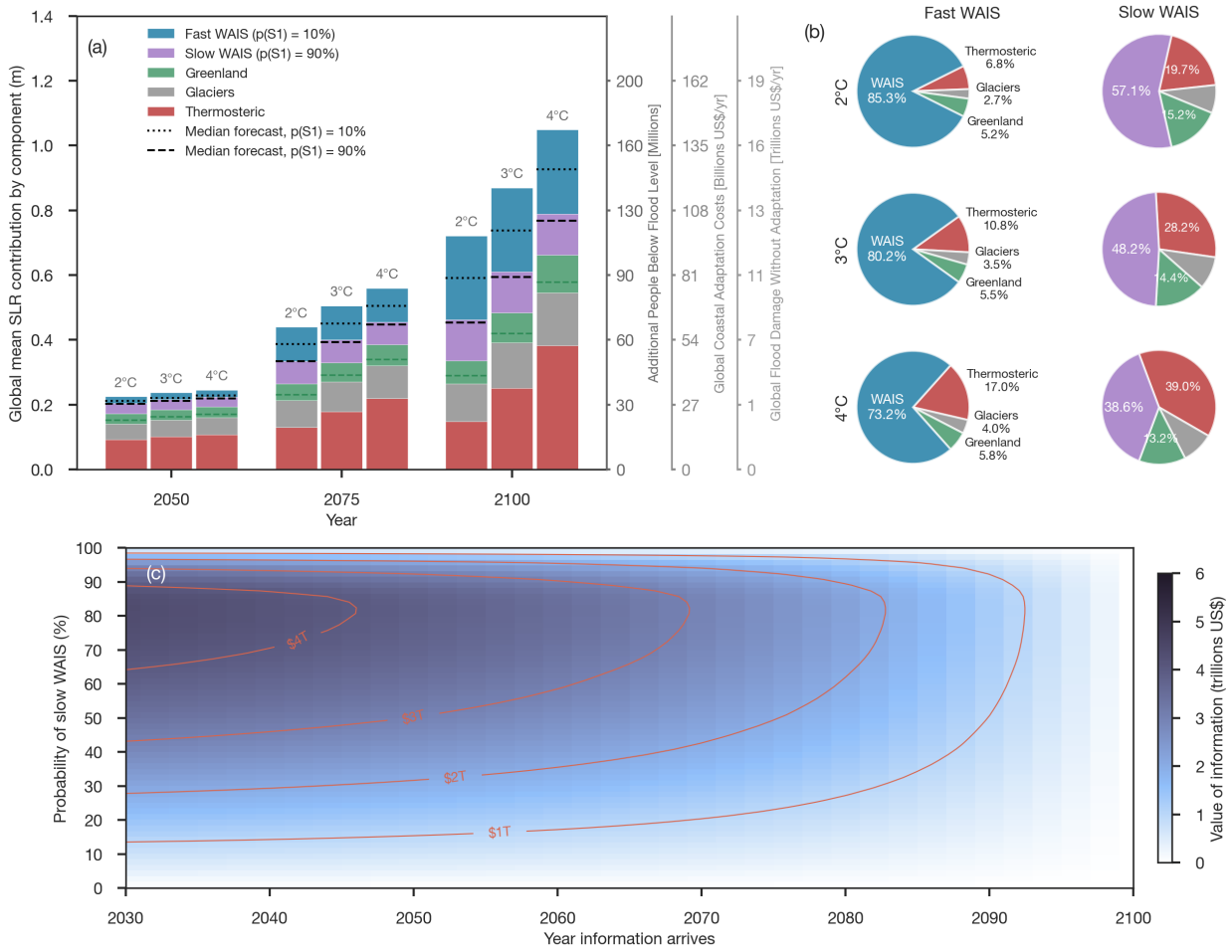


Figure 11: **Component contributions, variance decomposition, and value of information. (a)** 21st century mean GMSLR contribution by component and respective societal costs. Green dashed lines within Greenland bars partition discharge anomaly (below) from SMB anomaly (above). The purple box shows the WAIS contribution where slow mass loss is very likely, while the blue bar shows the additional potential SLR from rapid ice loss. The right-hand axes show previously published estimates of the additional number of people on land below flood level (*Kulp and Strauss*, 2019), global coastal adaptation costs (*Tiggeloven et al.*, 2020), and global flood damages without adaptation (*Jevrejeva et al.*, 2018) (Supplementary Material). **(b)** Shares of the 90% credible interval at 2100, taken as each component's own interval width as a fraction of the summed widths. **(c)** Estimated financial value of resolving WAIS uncertainty as a function of when information arrives. Contours and colormap in trillions US$ based on coastal adaptation costs (Supplemental Material). The plot shows that information has more value if it arrives early enough to inform key decisions and if it provides sufficient confidence that WAIS will lose mass slowly, allowing planning around lower future sea levels.